Supplement S2 Methods and Additional Results for Printed Polymer Context Recognition

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This supplement accompanies the article. It preserves the detailed protocol, full original outcomes and additional analyses while the main text focuses on material context recognition. Reference numbers follow the article. The original fixed-coefficient M2 benchmark and archived Simulink evidence are distinguished from the P2/M2P2 analysis, finite Walsh sensitivity and broader Walsh response in Sections S9–S10. Section S12 adds the matched uncoded and ordinary-coded taskwise comparison, retaining the same seven-profile Walsh results.

S1 Related study design and evidence scope

Table S1. Source and related application studies. Different sensing, targets and datasets preclude ranking their reported scores against the present recognition task.

Study and role	Observations and target	Learning and history	Evaluation focus	Relation to this study
Aktepe and Ergün [5] Source study	FDM process/material data; quality outputs	Hybrid ensemble prediction	Mechanical and physical quality	Source dataset; this study evaluates extracted record contexts.
Kim et al. [6] Material modelling	PLA/ABS microscopy classification; separate tensile-response regression	CNN image model; PCA/LSTM tensile model; no reported sequential class introduction	260 microscopy images; classification and stress-response modelling	Different inputs and targets; class percentages have unclear test denominators/aggregation and are not directly compared.
Trovato et al. [14] Defect recognition	Polymer powder-bed images; defect labels	Fixed image features + incremental ridge; summarized history	Accuracy and measured training resources	Different sensing and target; informs cost-aware updating.
Shen et al. [15] Process monitoring	Injection-molding process records; product-weight regression	Drift-triggered updates; accumulated old and new data	Quality monitoring under operating-condition drift	Observed production updating; different process, target and history access.
Subramanian et al. [16] Curve prediction	MEX laminate tensile curves; stress–strain regression	Anchors + residual updates; all accepted curves reused	Full-curve error and new-architecture validation	Sequential property prediction with accumulated history.
Present study Context recognition	104 source averages / 26 groups; two mechanical features	TaskIL/ClassIL; explicit current, limited, summarized or full history	Whole-group folds; acquisition/retention; coverage/work	Exploratory benchmark of source-record procedures.

Kim et al. [6] report distinct microscopy-classification and tensile-response modelling tasks. These differ from recognition using modulus and hardness under the present class-introduction protocol. Trovato et al. [14] include a simple incremental ridge readout. The present comparison therefore focuses on a specific mechanical-input task with stored-record and centroid references.

S2 Exact training and mathematical specification

All trained methods in the original fixed-coefficient benchmark use seeds 101, 211, 307, 401 and 503, Xavier-uniform weights, zero biases and binary64 arithmetic. Networks of the same family begin with identical weights for a given seed. Every stage has 600 full-batch Adam updates, learning rate 0.02, beta values 0.9 and 0.999, and epsilon 1e-8 outside the square root. Weight-only L2 regularization is one half of 1e-4 times the sum of squared weight entries; biases are excluded. Sequential methods carry their own optimizer state. Joint models reset weights and moments to the same initialization at every stage.

Equal update counts do not imply equal computation: replay, distillation and Fisher estimation require additional operations. The common small network, full-batch updates and fixed coefficients are explicit

adaptations for this experiment. They are not asserted to be the original papers' default settings. In that original benchmark, no coefficient or seed is chosen from its held-out evaluation outcomes. The added finite inner-development selection is described separately in Section S9.

Zhou et al. [22] motivate accounting for both model and exemplar memory. We report these components separately; the comparison retains different total memory and model capacities.

At the first arrival of a class, replay stores two current training rows. Selection is the first two entries of a fixed SHA-256 ordering of fold, class and source-row identity. Features, prediction errors and held-out data do not enter selection. The same rows are used by all relevant methods and seeds. Previous-class rows are neither rescanned nor reselected. The post-stage buffer contains 4, 6 and 8 unique rows.

Rows already in the current batch are removed from the memory loss. Effective replay therefore contributes 0/2/4 rows in TaskIL and 0/4/6 in ClassIL. When memory is nonempty, current and memory mean losses each receive weight one half. Otherwise only the current mean loss is used. A TaskIL memory row retains its first-admission pair mask. Coded memory targets remain the fixed true class code, including in negotiated arms. This deterministic replay adaptation implements the replay principle [11, 13] without claiming a verbatim reproduction of a source benchmark's random sampling.

For each completed stage r, EWC stores its final parameter vector and a diagonal model-Fisher vector. The Fisher averages squared individual log-probability gradients over that stage's training inputs and all eligible classes, weighted by the model probabilities. It is neither an observed-label empirical Fisher nor the square of a batch-average gradient. In the following expression, j indexes parameters, i inputs and k eligible classes; N_r is the number of current training inputs and K_r the eligible set.

$$F_{rj} = \frac{1}{N_r} \sum_{i=1}^{N_r} \sum_{k \in K_r} p_{\theta_r}(k \mid x_i) \left(\frac{\partial \log p_{\theta_r}(k \mid x_i)}{\partial \theta_j} \right)^2$$

The current supervised objective and L2 term are supplemented by the following importance penalty. Each earlier stage retains its own parameter anchor. Stages 1 and 2 provide the two Fisher/anchor pairs used at stage 3; the unused final Fisher is not computed. Regularization and EWC derivatives do not enter Fisher estimation.

$$L_{EWC} = \frac{\lambda_{EWC}}{2} \sum_{r < s} \sum_{j} F_{rj} \left(\theta_j - \theta_{rj} \right)^2$$

The primary EWC coefficient is 10; coefficients 1 and 100 are reported separately as fixed sensitivities. TaskIL Fisher uses the current pair, and ClassIL Fisher uses all introduced classes even when the arriving batch has a single new label. These choices keep the likelihood support explicit and avoid a degenerate single-class softmax. The primary coefficient remains unchanged after evaluation.

LwF freezes the same arm's pre-stage network [19]. It evaluates current training inputs once at temperature 2 and caches the old-class response targets for the complete stage. Distillation cross entropy has coefficient 1 and no additional temperature-squared multiplier. TaskIL sums a loss over every earlier complete pair, rather than only the common PLA+ anchor. ClassIL uses the union of earlier classes. The current supervised loss, distillation loss and L2 term are optimized together. No earlier raw rows or Fisher state are accessed. This retains information in the declared old-label masks; it does not automatically calibrate old and new probability masses.

Ordinary coding uses the four rows of [I4, 0] in eight bits. Walsh coding uses rows 2–5 of the binary transformation of the Sylvester matrix below. The dictionaries and class order PLA+, ABS, PETG, PLA-CF are fixed before learning. Each ordinary row has one active bit; each Walsh row has four. The comparison concerns the complete codebook geometry, including bit density, and does not isolate orthogonality.

$$H_1 = [1]; H_{2n} = \begin{bmatrix} H_n & H_n \\ H_n & -H_n \end{bmatrix}; \quad C = \frac{1 - H_8(2{:}5, :)}{2}$$

Negotiated representations are attributed to the explicitly identified preprint by Korhan and Öner [20] and its accompanying implementation. Our adaptation fixes the class dictionary and decoder, restricts teacher access to current training inputs, and crosses negotiation with the declared replay rule. The stage-start network supplies the following mixed targets, where sigma is the component-wise sigmoid and x_i is the standardized input. Targets are cached once and held fixed during all 600 updates.

$$q_{si}=(1-\alpha_s)C_{y_i}+\alpha_s\sigma\left(f_{\theta_{s-1}}(x_i)\right);(\alpha_1,\alpha_2,\alpha_3)=\left(\frac{1}{2},\frac{2}{3},\frac{3}{4}\right)$$

At stage 1 the teacher is the untrained initialization. Its contribution is target smoothing, not preservation of previously learned knowledge. Static-target controls set every alpha to zero. Current targets use mean binary cross entropy over eight bits; replay uses hard class-code targets. The displayed objective is the current-data term plus weight regularization. When memory is present, the two data means receive the one-half weights described above. Here z denotes a pre-sigmoid output and W1/W2 the two weight matrices.

$$L_s=\frac{1}{N_s}\sum_{i=1}^{N_s}\frac{1}{8}\sum_{b=1}^{8}\left\{\log(1+\exp z_{ib})-q_{sib}z_{ib}\right\}+\frac{\lambda}{2}\left(\|W_1\|_F^2+\|W_2\|_F^2\right)$$

For a sigmoid output vector p, prediction selects the eligible class with the smallest mean squared code distance:

$$d_k(p)=\frac{1}{8}\sum_{b=1}^{8}(p_b-C_{kb})^2$$

The code gap is the second-smallest minus smallest eligible distance. Cross-entropy decoders use the largest eligible softmax score and the top-two score difference. PROTO and NN1_FULL use their squared-distance separation. Exact ties are flagged and resolved by the lowest global eligible label. No approximate tie tolerance changes the selected class. These different gaps are descriptive native scores, not calibrated probabilities on a common scale.

The primary outcome is final ClassIL class-balanced forced accuracy. Counts are combined across held-out groups within each seed before calculating rates, rather than averaging five unequal folds equally. Each class contributes 26 final rows. TaskIL reports the macro average over three equally sized tasks, giving 156 final views of 104 unique rows. Current and old-unit counts are reported at every stage. Invalid decisions remain in the denominator and count as wrong; nonfinite model or prediction outputs also prevent technical acceptance.

Acquisition is accuracy when a task or class first arrives. Backward transfer (BWT) is final minus acquisition accuracy averaged equally across the old units: tasks 1/2 or classes 1/2/3. Signed forgetting is each old unit's best earlier post-acquisition accuracy minus its final accuracy, averaged over the same units. It is not clipped at zero. The initial stage has no old-unit value. These measures are read together with actual acquisition and final old/new counts, because weaker acquisition can produce a small subsequent decline.

Coverage curves are constructed separately for each fold, seed, arm, stream and stage. Thresholds are positive infinity and the distinct finite gaps in decreasing order. Every exact-gap block enters together, using gap greater than or equal to threshold. For the fixed population N, C is correctly answered, W incorrectly answered, A valid but unanswered, and F invalid; N = C + W + A + F. Coverage is (C + W)/N and conditional error is W/(C + W), undefined when no query is answered. Labels evaluate error but do not choose threshold values or an operating point. Raw scores from different folds are not pooled. The original fixed-allocation 0.15 rule is not transferred to this benchmark.

All five seeds and the minimum, median and maximum are retained. We report descriptive matched differences, confusion matrices and separate native coverage curves. No p-values, specimen-level confidence intervals or best-seed result are inferred from folds or optimizer starts. Persistent model, normalization, class-state, replay, teacher, target, Fisher and optimizer arrays are counted explicitly. Numerical payload bytes,

saved-file size and process commit are distinct quantities. Joint-training work also includes the raw history it requires.

The MATLAB implementation saves initial/final weights, Adam states, teacher and target caches, Fisher vectors, access memberships and every prediction. Independent calculations check first/last checkpoint losses, gradients and Adam steps, all Fisher jobs and saved inference. The intervening 598 updates are not independently retrained. This is implementation consistency evidence within the source-record experiment. Exact aliases share only jobs with identical inputs, targets, incoming network/optimizer state and specification. Standalone logical work remains fully charged to each method.

S3 Complete original M2 result vectors

Table S2. Final simple-reference correct counts. Linear vectors follow seeds 101, 211, 307, 401 and 503; deterministic rules appear once. Counts pool held-out groups before computing rates. NN1_FULL retains all introduced training records.

Reference	TaskIL correct / 156	ClassIL correct / 104
PROTO	128 / 156	66 / 104
NN1_FULL	138 / 156	66 / 104
LIN_SEQ	121, 120, 120, 122, 121 / 156	26, 26, 26, 26, 26 / 104
LIN_JOINT_GLOBAL	125, 125, 125, 125, 124 / 156	66, 66, 66, 66, 65 / 104

Table S3. Final matched correct-count vectors followed by median accuracy. TaskIL denominator 156; ClassIL denominator 104. Vectors retain seeds 101/211/307/401/503. The final two rows are EWC sensitivity arms, not replacements for primary coefficient 10. Minimum/maximum and every class/task count are supplied in Supplement S1.

Method	TaskIL vector; median	ClassIL vector; median
CE naive	112, 117, 115, 114, 107; 73.08%	26, 26, 26, 26, 28; 25.00%
CE replay	112, 114, 117, 122, 124; 75.00%	61, 62, 57, 60, 56; 57.69%
CE EWC 10	129, 135, 134, 128, 136; 85.90%	32, 31, 31, 31, 32; 29.81%
CE LwF	130, 128, 127, 133, 128; 82.05%	29, 26, 28, 25, 27; 25.96%
CE joint global	136, 131, 134, 133, 135; 85.90%	69, 68, 69, 68, 69; 66.35%
Ordinary static	95, 95, 94, 96, 101; 60.90%	26, 26, 26, 26, 26; 25.00%
Ordinary static + replay	129, 124, 126, 125, 128; 80.77%	63, 64, 66, 69, 62; 61.54%
Ordinary negotiated	120, 114, 114, 116, 115; 73.72%	27, 43, 42, 32, 37; 35.58%
Ordinary negotiated + replay	125, 123, 122, 122, 124; 78.85%	51, 49, 46, 49, 53; 47.12%
Walsh static	94, 97, 95, 95, 100; 60.90%	26, 26, 26, 26, 26; 25.00%
Walsh static + replay	120, 121, 122, 121, 122; 77.56%	70, 64, 68, 67, 65; 64.42%
Walsh negotiated	114, 119, 115, 117, 116; 74.36%	23, 39, 37, 34, 34; 32.69%
Walsh negotiated + replay	126, 124, 123, 129, 122; 79.49%	52, 47, 48, 49, 50; 47.12%
CE EWC 1 (sensitivity)	133, 135, 123, 129, 131; 83.97%	32, 33, 28, 35, 33; 31.73%
CE EWC 100 (sensitivity)	134, 136, 136, 132, 135; 86.54%	28, 30, 29, 27, 31; 27.88%

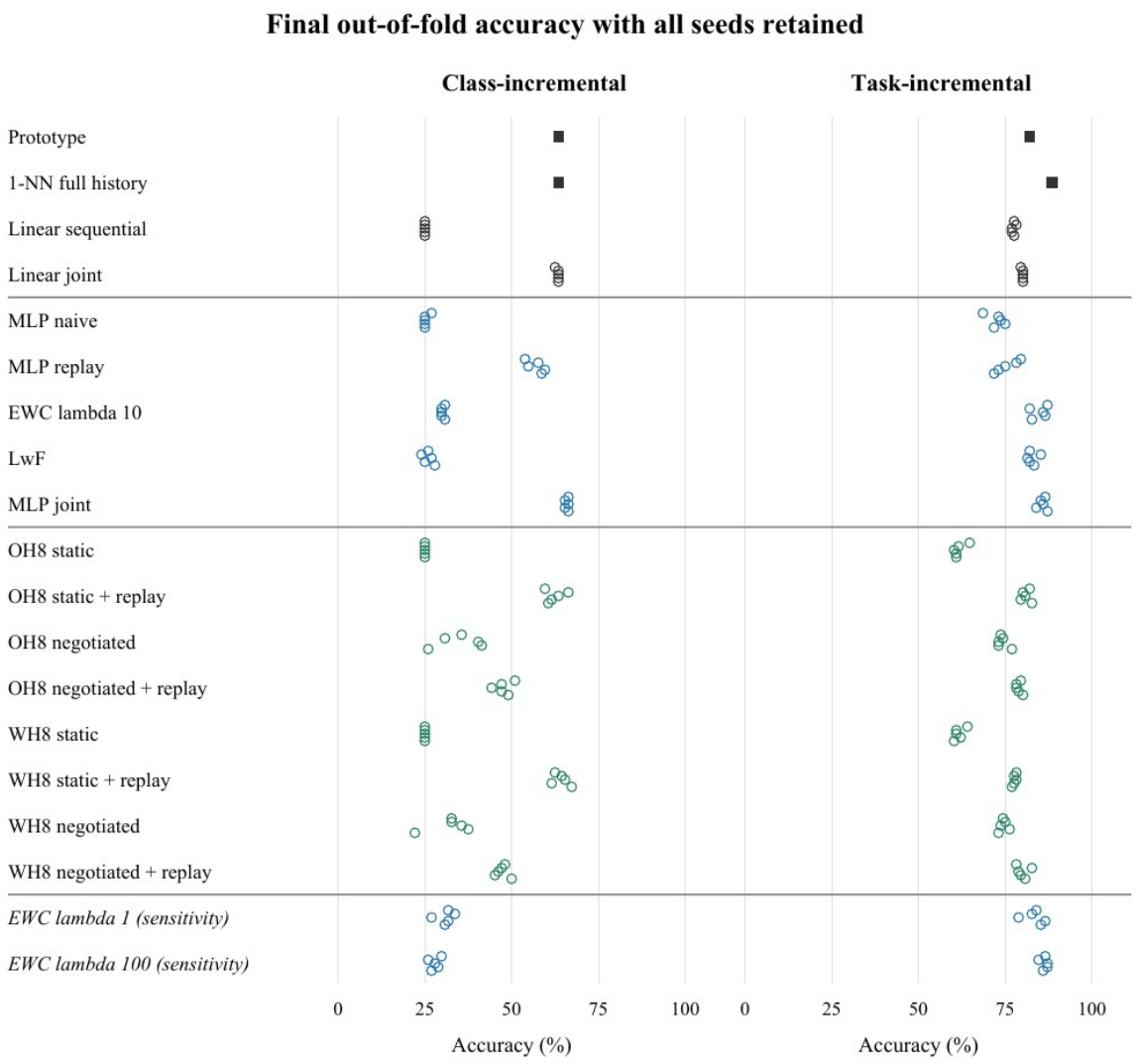


Figure S1. Final held-out-group forced accuracy for all trained arms and the four simple references. Each trained-method marker is one of the five fixed seeds; deterministic references appear once. Each seed combines the same 26 held-out condition groups. EWC coefficients 1 and 100 are labelled sensitivities, while coefficient 10 remains primary. The CE and coded blocks differ in capacity, objective and decoder. TaskIL and ClassIL retain their own candidate information and denominators.

S4 Coverage and computational accounts

Forced decisions give one operating condition: every valid query is answered. Supplement S1 also provides 2250 separate native-gap coverage curves for the matched block, retaining every fold, seed, method, stream and stage. Figure S2 shows final-stage curves for replay MLP, primary EWC and negotiated Walsh with replay. Every plotted point is a realizable whole-gap block. Conditional error may increase or decrease as more queries are answered; no curve is forced to be monotone and no threshold is selected from its error labels.

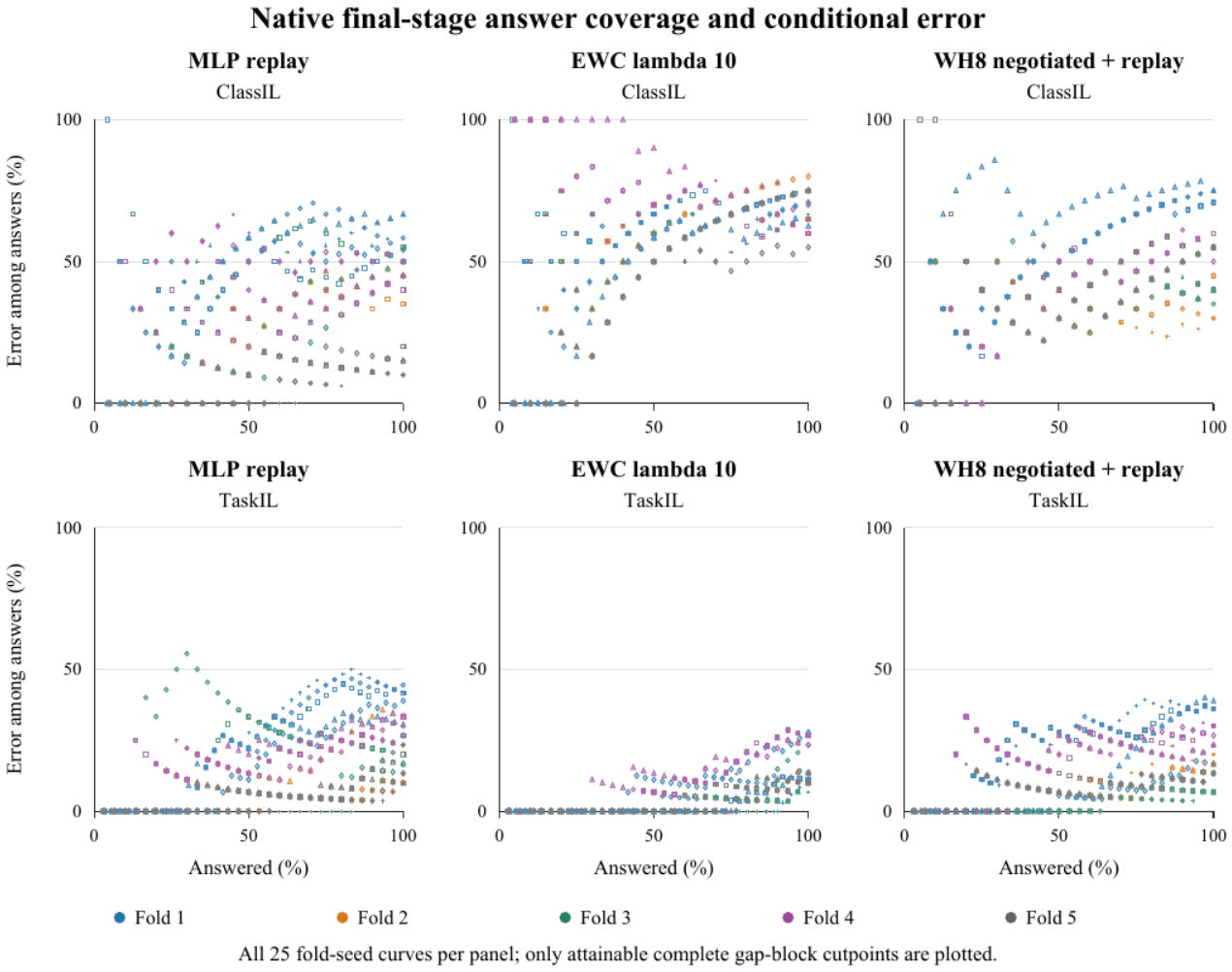


Figure S2. Native-score answer coverage and conditional error at the final stage. Each panel retains all 25 fold/seed sets of attainable cutpoints for one method and stream. Every marker represents a complete equal-gap block; no interpolated operating thresholds are displayed. Fold 1 has six condition groups and the others five. The zero-answer point has undefined conditional error and is not plotted as zero error. Fold colours and seed marker shapes preserve the separate evaluations. Raw gaps are never pooled between folds or score families; the points imply no calibration or risk guarantee.

Coverage is useful because a ranking can be informative at one answer rate while its uncertain decisions remain visible at another. Here a user-level loss for withholding versus answering incorrectly has not been specified. We consequently report the complete answer/error trade-off rather than nominate a universally preferable threshold or method. The original 0.15 selected-answer result remains only in Section S6.

The final numerical prediction state is 136 bytes for PROTO, 1960–2056 for NN1_FULL, and 136 for the linear predictors. Persistent linear training state is 336 bytes for LIN_SEQ and 2256–2352 for LIN_JOINT_GLOBAL when its retained raw history is included. These figures include the declared normalization and class-state arrays. They exclude language/runtime objects, audit identifiers and temporary workspace. The nearest-record reference therefore offers strong accuracy with an explicitly larger raw-history state than PROTO. Table S4 summarizes the matched training-work counts.

Table S4. Standalone work per seed over all five folds. Each row has 9000 logical optimizer updates. Paired entries are TaskIL / ClassIL. Data rows include current plus effective replay; knowledge distillation (KD) gradient rows and teacher forwards are separate. Fisher counts are class-gradient terms. Ordinary and Walsh have the same declared workload for each target/replay setting. EWC sensitivities have the primary arm's work counts. Counts are computational exposures, not physical observations.

Method	Params	Data gradient rows	KD gradient rows	Teacher rows	Fisher terms
CE naive	32	374400 / 249600	0 / 0	0 / 0	0 / 0
CE replay	32	392400 / 279600	0 / 0	0 / 0	0 / 0
CE EWC 10	32	374400 / 249600	0 / 0	0 / 0	832 / 728
CE LwF	32	374400 / 249600	374400 / 124800	416 / 208	0 / 0
CE joint global	32	561600 / 561600	0 / 0	0 / 0	0 / 0
Either code static	52	374400 / 249600	0 / 0	0 / 0	0 / 0
Either code static + replay	52	392400 / 279600	0 / 0	0 / 0	0 / 0
Either code negotiated	52	374400 / 249600	0 / 0	624 / 416	0 / 0
Either code negotiated + replay	52	392400 / 279600	0 / 0	624 / 416	0 / 0

For each matched arm and seed, all five folds account for 9000 logical updates. For sequential TaskIL learners, current-data gradient rows total 374400, rising by 18000 memory rows under replay. Sequential ClassIL current rows total 249600, rising by 30000 under replay. Global joint MLP instead processes 561600 current/history gradient rows in either stream. LwF additionally processes 374400 TaskIL or 124800 ClassIL distillation-gradient rows. Its teacher-forward counts are 416 and 208. Negotiated coded targets require 624 and 416 stage-start teacher rows. These are computational exposures, not additional material tests.

Primary EWC adds 416/312 Fisher input rows and 832/728 class-gradient terms per seed across the five TaskIL/ClassIL folds. The two saved anchor/Fisher pairs add at most 1024 numerical bytes. LwF separately retains a 256-byte teacher and up to 1344/504 bytes of cached TaskIL/ClassIL targets. Code learners store 416 bytes of weights and a 256-byte codebook, with an 840-byte Adam state; negotiated teachers and target caches are additional. Equal parameter counts within a block do not make these historical-information costs equal.

Across the actual shared-job producer, the five matched processes completed in 181.655 seconds total, with a maximum owned-process commit of 1745461248 bytes. Each process stayed within its 600-second, 64-MiB output and 4-GiB commit limits and closed its owned job. This is an aggregate implementation measurement, not per-method inference latency, peak model memory, energy or a real-time deployment guarantee. Standalone logical costs remain separate from shared execution savings.

These supporting accounts describe a trade-off rather than a single utility winner. The compact prototype and full-history nearest rule define useful endpoints for state and accuracy. More elaborate procedures add different teacher, Fisher and replay work, and their benefits depend on candidate information. Separate coverage curves preserve a further decision dimension until an application specifies the relative costs of errors and withheld answers.

S5 Archived M2 frozen inference checks

The historical fixed-allocation model bank and its two Simulink demonstrations are described in Section S6. The following checks concern the original M2 five-fold bank.

Five additional self-contained Simulink models implement the saved five-fold bank. They cover all stages, applicable seeds and logical model aliases in both streams, including the four simple references and all 15 neural or coded arms. Starting from the two raw measurements, the models apply the frozen fold normalizer and reproduce 47,502 query views across 2,610 logical model-stage populations.

Independent checks covered all 35 output channels, including normalized inputs, decoder values, candidate masks, labels, ties, validity and score gaps. Numerical agreement used an absolute tolerance of 2×10^{-11} plus 2

$\times 10^{-11}$ times the absolute reference value. The largest observed absolute discrepancy was 9.77×10^{-15}. Native gap-block membership and order, and every C/W/A/F/N count, matched exactly. These checks verify frozen inference; they add no training, physical response model or specimen validation.

These archived checks apply to the original M2 models. The additional P2/M2P2 references retain their separate recorded numerical analysis. Section S12.4 reports the original-reference verification of the matched seven-profile NL4/OH8/WH8 bank, including exact confidence-gap group membership and coverage counts. Earlier sensitivity trajectories retain their separately stated scope. Learning used MATLAB R2026a on Windows. The saved-state and Simulink comparisons establish consistency of frozen inference.

S6 Historical fixed allocation

This section preserves the historical fixed-allocation source-record investigation. Its allocation has 72 TRAIN, 16 first-development (FIT) and 16 reporting-development (REPORT) averages in 18/4/4 condition groups. All records were previously inspected. Final TaskIL reporting comprises 24 views of 16 unique averages in four groups. These outcomes remain separate from the five-fold benchmark in the main text.

In the original score-based arm, replay stores the first two training records by frozen group order for each class of a completed task. It retains the original task masks. Current and memory mean-loss masses are 1 and 0.5, normalized by their total. Selected answers use a fixed pair-score gap of 0.15, alongside a separate forced decision. Table S5 retains the original reporting outcomes. These rules differ from the hash-selected, deduplicated, equal-mass replay and native coverage curves of the new benchmark.

Table S5. Original fixed-allocation final REPORT decision counts. Entries are minimum / median / maximum across five seeds, or one invariant value. Each run has 24 TaskIL views of 16 unique published averages in four groups. Selected answers use the original 0.15 gap; forced decisions are separate. This population is not pooled with the five-fold benchmark.

Original method	Selected correct	Selected wrong	Unanswered	Forced correct
Linear, no replay	19	1 / 1 / 2	3 / 4 / 4	21
Linear, replay	20 / 21 / 21	0 / 0 / 1	3	22 / 23 / 23
Linear, joint seen	18 / 19 / 19	1	4 / 4 / 5	21
MLP, no replay	19 / 21 / 22	2 / 3 / 5	0 / 0 / 1	19 / 21 / 22
MLP, replay	23 / 24 / 24	0 / 0 / 1	0	23 / 24 / 24
MLP, joint seen	24	0	0	24
Nearest seen	19	0	5	24

Original neural replay gives 23–24 correct selected REPORT views and gains 2–4 old-task forced decisions relative to no replay in every paired seed. In FIT, the old-task forced gains are +1, 0, −1, 0, −1. The original newest-task FIT acquisition marker is six correct of eight; replay meets it in two of five seeds. Retaining this contrast documents the dependence of the original result on its small fixed condition populations.

The original code comparison has no historical-example replay in the four code arms. Negotiated Walsh gives 17–20/24 forced REPORT TaskIL decisions versus 15–16 for its static control. It retains 10–13/16 old-task views and gives 7/8 new-task views. In the original ClassIL reporting population, negotiated Walsh gives 4–6/16, with 4/12 old and 0–2/4 new records correct. CE replay gives 11–13/16, with 8–9/12 old and 2–4/4 new records correct. These fixed-allocation outcomes are preserved without treating them as extra held-out groups in the new analysis.

Two existing Simulink models reproduce 30 score-based and 55 coded/class-stream final states on 720 and 1040 decision views. Their recorded maximum absolute differences from the numerical references are 5.33×10^{-15} and 8.88×10^{-16}. The declared tolerance is $10^{-10} + 10^{-10}$ times the absolute reference value, with exact eligible-mask, label and tie agreement. These original-bank checks concern frozen inference consistency, not training in Simulink or physical response models. The five-fold verification is reported separately in the main text. The original figure sources and model files remain available in the v27.0 archive.

S7 Pairwise confusions and additional feature views

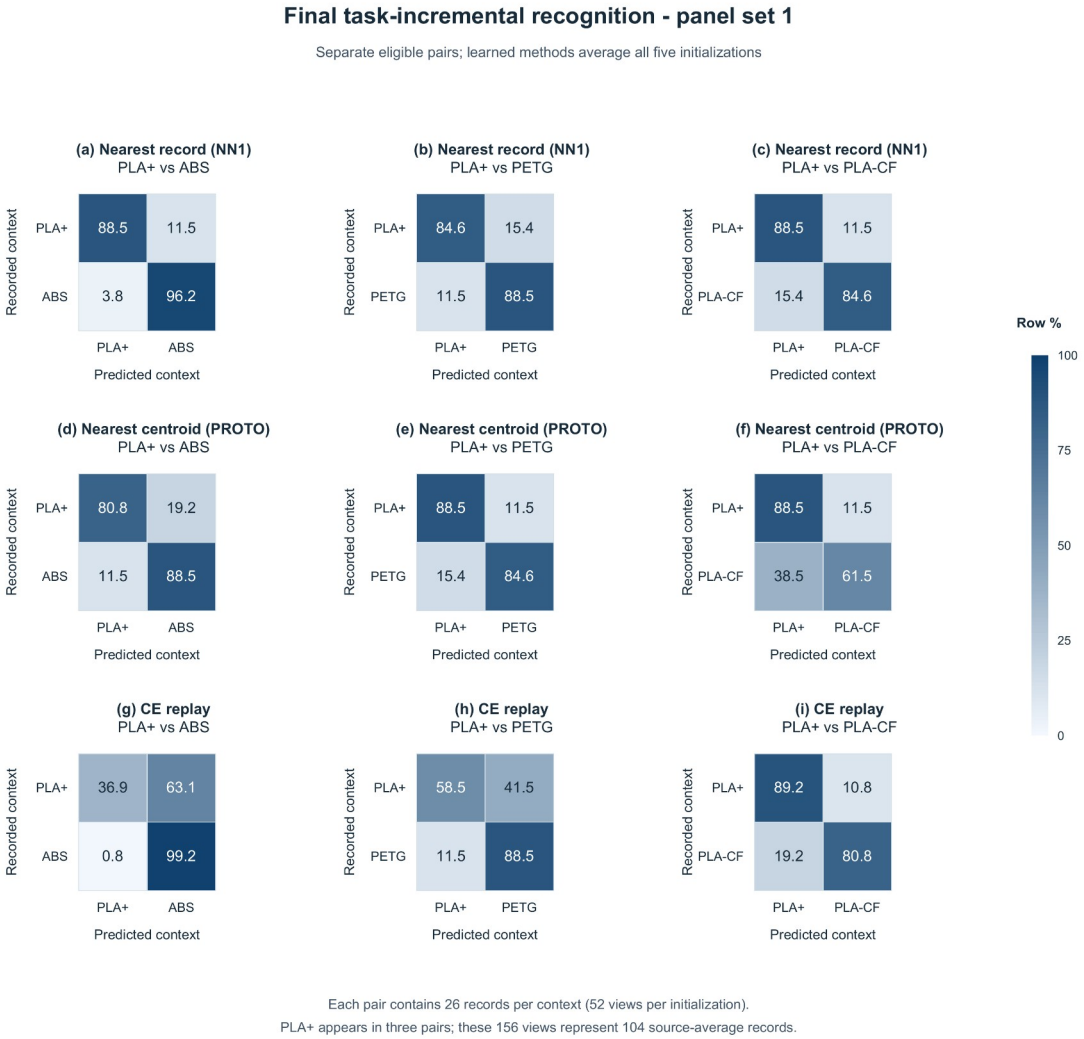


Figure S3. Final TaskIL confusion matrices for nearest record (a–c), centroid (d–f) and CE replay (g–i), under PLA+/ABS, PLA+/PETG and PLA+/PLA-CF pairs. Each row contains 26 held-out average records. Values are row percentages; CE replay averages all five starts. PLA+ is reused across the three pairs, so 156 decision views represent 104 source averages.

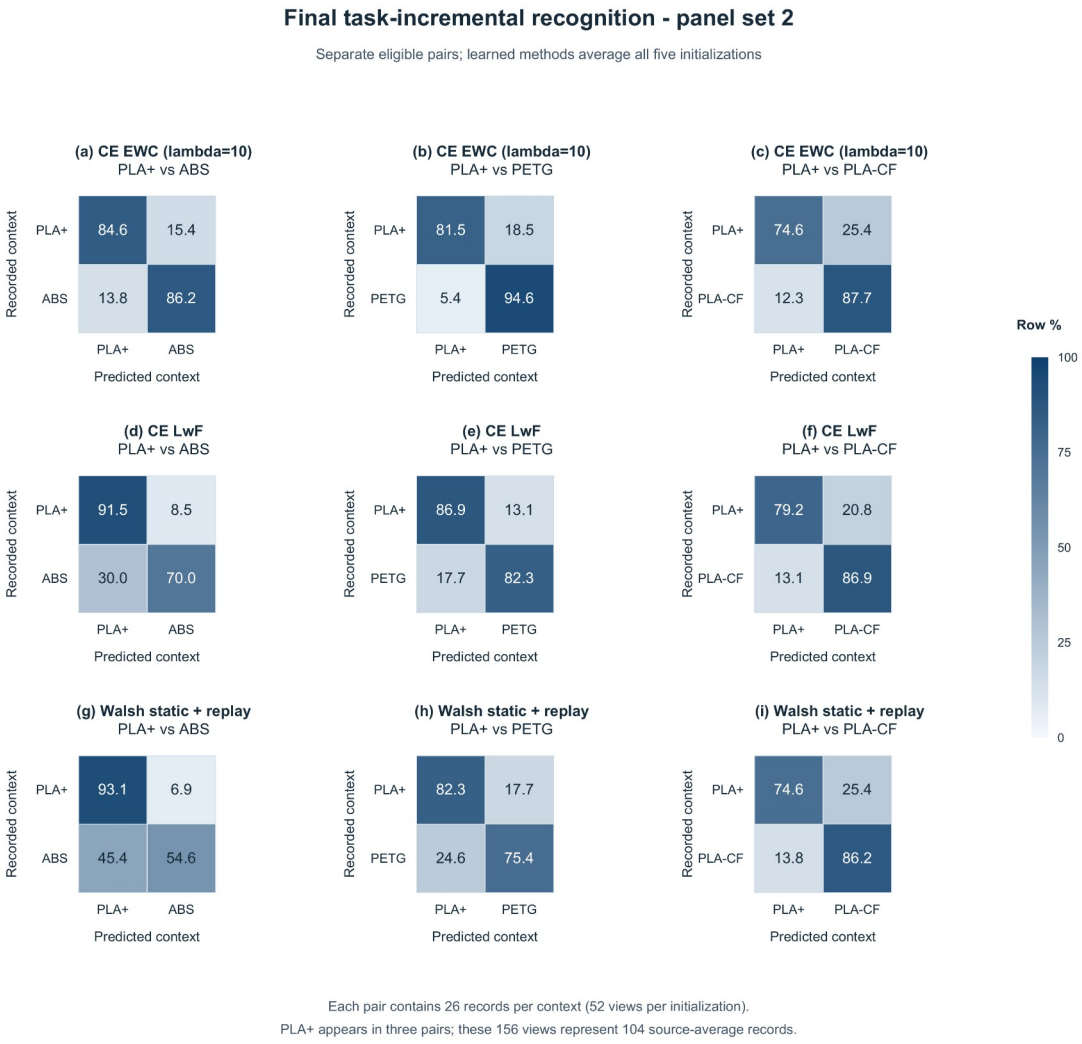


Figure S4. Final TaskIL confusions for primary EWC (a–c), LwF (d–f) and Walsh static with replay (g–i), using the same three pairs as Figure S3. Values average row percentages over all five starts, with 26 source averages per row and start. Pair identities are supplied equally to all methods. These pair-conditioned results do not represent unrestricted four-class recognition.



Figure S5. Material-faceted view of the same measured feature space, coloured by infill. Every facet contains the 26 source averages for the named context. Coordinates, linear axes and limits match main Figure 2. Nearby marks retain their original coordinates; the accompanying table preserves point identities and verified metadata.

S8 Printing-input implementation and complete records

The fixed-fold comparison uses P2 [layer_mm, infill_percent] and M2P2 [E_MPa, Hardness_ShoreD, layer_mm, infill_percent]. M2 predictions are reused byte-identically from the accepted archive. Exact row/group/label/source-Excel joins admit all 104 records. Process scales use only the initial TRAIN classes; old M2 scalar scales and fit identities remain exact. PROTO sums/counts and NN1 history admit each introduced TRAIN row once, in canonical group/label order.

One bounded MATLAB execution saved 2,184 new stage-query rows, including 1,040 final rows, 30 shared feature/fold/stage states and 10 normalizers. The 1,092 original deterministic M2 views were reused without forward rerunning; the three-feature-set final comparison contains 1,560 rows. Source-free controls passed 44 checks. A separate saved-output review passed 101,549 checks with zero failures, covering every new candidate distance, label, tie, gap, source identity, scaler, history state and aggregate. This does not add specimens or a physical experiment.

The full final M2P2-vs-M2 paired records include 208 method/stream/feature/group summaries, 32 class summaries, 40 fold summaries and 1,040 individual final query comparisons across the two added-input feature sets. For NN1's M2P2 TaskIL results, 11 groups gain correct decisions, four lose and 11 retain the

same total. ClassIL has 16 gaining groups, three losing and seven unchanged. ClassIL has 34 changed predictions: 25 corrections, seven regressions and two changes between wrong labels. Correctness changes and prediction changes are distinct.

For every class, the process training vectors have identical support. PROTO's added distance term is therefore common to eligible labels in exact arithmetic. The prospective expectation was checked against all 546 augmented PROTO views: no labels or exact tie states changed; maximum additive residual was $7.105427357601002 \times 10^{-15}$. P2 alone produced exact ties, resolved by the smallest eligible label. No epsilon, feature weight, threshold, seed or fold was tuned to obtain these results.

The printing-input files in this revision retain the admitted 104-row metadata table, original fold memberships, frozen old scaler copies, prospective protocol and mathematical note, new MATLAB reference implementation, complete saved states/predictions, paired summaries and independent audit receipts. Existing two-input CL outcomes remain the common-information benchmark. All source means were previously exposed, and process variables are observational metadata rather than isolated experimental interventions.

S9 Finite negotiated-target sensitivity

Design, visible information and candidate family

This separately frozen sensitivity retains M2 inputs, the original five outer whole-condition folds, the 2–4–8 Walsh network and its 52 parameters, replay absent/present, and the original optimizer and decoder. Alpha is dimensionless target mixing; Adam eta remains 0.02. For alpha greater than zero, the same arm's incoming network evaluates only current TRAIN inputs once before that stage. Its mixed targets are detached and cached for all 600 updates. Stage 1 otherwise uses an untrained network, so its role is target smoothing. At alpha zero, the hard code is used exactly and no teacher forward or soft teacher-target cache is required. The all-static procedure retains model and optimizer state but no negotiated teacher state or soft teacher targets. Replay always uses hard codes for its admitted old rows.

Table S6 lists the eleven declared coarse vectors and the two possible midpoints actually generated. Coefficients change only between the three stages. This includes increasing-then-decreasing and decreasing-then-increasing vectors, but does not implement a within-update or state-adaptive rate. The ordinary dictionary and other learning families were not re-searched.

Table S6. Three-stage alpha vectors. R1 and R2 were generated only for the no-replay contexts of outer folds 1 and 2, after the predeclared screen-count trigger. The 2/3 entries are consumed as their binary64 representation. IDs identify prospective vectors, not claims of optimality.

Candidate ID	Stage 1	Stage 2	Stage 3
F00_STATIC	0	0	0
F25	0.25	0.25	0.25
F50	0.5	0.5	0.5
F75	0.75	0.75	0.75
F90	0.9	0.9	0.9
UP_NR_LEGACY	0.5	2/3	0.75
UP_LINEAR	0.25	0.5	0.75
DOWN_LINEAR	0.75	0.5	0.25
HARD_FIRST_THEN_NR	0	2/3	0.75
PEAK	0.25	0.75	0.25
VALLEY	0.75	0.25	0.75
R1_CONST	0.625	0.625	0.625
R2_MIX	0.25	2/3	0.75

Within each outer training pool, screen2 and confirm3 assign intact groups to two or three development folds using a fixed SHA-256 ordering independent of measurements and outcomes. Each group contains all four labels. Every available group appears exactly once in the DEV union of that design. Its pooled final ClassIL denominator is 80 for outer fold 1 and 84 for the other folds, with 20 or 21 rows per class. Each inner scaler uses only its own initial TRAIN PLA+/ABS rows and remains fixed; final outer trajectories reuse the accepted outer scaler. DEV outputs do not modify fitting, targets, buffer, stopping or model state. Outer query outcomes do not generate candidates or choose vectors.

Screening evaluates all eleven vectors with starts 101 and 211. Candidate counts are pooled across inner groups within each seed before ranking. The descending primary key is median total correct; ties prefer the larger minimum seed count, then a constant vector, lower mean alpha and lexical ASCII ID. The doubled median is an integer, avoiding percentage-rounding decisions. Refinement requires the same unique raw-count winner and a lead of at least two correct decisions over the runner-up in each screen seed. This is a prospective resolution rule, not a significance test.

When the trigger holds, R1 is the first novel midpoint between the highest-ranked constant and its immediate constant neighbour, choosing the better-ranked neighbour and then the lower coefficient/ID for ties. R2 is the first novel nonconstant midpoint from coarse-rank pairs ordered by rank-index sum, maximum index and individual indices, excluding two-constant pairs. Duplicate vectors are skipped by exact binary64 identity. Each context receives at most one R1 and one R2; there is no recursive expansion. The same two screen splits and two starts evaluate them. Only folds 1 and 2 without replay qualified, with winner margins 7/2 and 9/2 correct in the two starts. Both produced R1=(0.625,0.625,0.625) from F75/F50 and R2=(0.25,2/3,0.75) from hard-first/legacy.

The top two complete-screen candidates per context then receive fresh-initialization confirmation on confirm3 with all five original starts. Confirmation counts alone choose the winner under the same ranking rule; screen counts are not added to confirmation counts. The ten vectors are frozen before any final outer query access. Each selected vector is then used in a ClassIL trajectory and an independently trained TaskIL trajectory with its own pair arrivals. Thus TaskIL checks transfer of a ClassIL-selected coefficient; it is not separately tuned. Table S7 gives the choices, and Tables S11–S12 retain all development candidates and count vectors.

Table S7. Confirmed vector for every outer-fold/replay context. One choice applies to all five final starts in that context and to both separately trained streams. Alpha zero means static hard targets, with no negotiated teacher state or soft teacher targets.

Outer fold	Replay	Selected candidate	Alpha vector
1	No	UP_NR_LEGACY	(0.5, 2/3, 0.75)
1	Yes	F00_STATIC	(0, 0, 0)
2	No	HARD_FIRST_THEN_NR	(0, 2/3, 0.75)
2	Yes	F00_STATIC	(0, 0, 0)
3	No	HARD_FIRST_THEN_NR	(0, 2/3, 0.75)
3	Yes	PEAK	(0.25, 0.75, 0.25)
4	No	HARD_FIRST_THEN_NR	(0, 2/3, 0.75)
4	Yes	F25	(0.25, 0.25, 0.25)
5	No	HARD_FIRST_THEN_NR	(0, 2/3, 0.75)
5	Yes	F00_STATIC	(0, 0, 0)

Final outcomes and acquisition

Table S8. All final outer summaries. Counts follow starts 101/211/307/401/503; deterministic NN1 appears once. Accuracy is median [minimum–maximum] percent. Final denominators are 104 ClassIL or 156 TaskIL views; the latter reuses 104 source averages across pairs. Static and legacy rows are the original pinned results, not routine retraining. Selected rows are the complete inner-selected procedure. All outcomes remain exposed-source, exploratory evaluations.

Procedure / history	Stream	Correct counts	Accuracy (%)
Inner-selected WH8, no replay	ClassIL	38, 43, 43, 40, 43	41.35 [36.54–41.35]
Inner-selected WH8, no replay	TaskIL	110, 111, 113, 115, 112	71.79 [70.51–73.72]
Inner-selected WH8 + replay	ClassIL	61, 60, 66, 57, 59	57.69 [54.81–63.46]
Inner-selected WH8 + replay	TaskIL	119, 121, 123, 120, 121	77.56 [76.28–78.85]
Static WH8, no replay	TaskIL	94, 97, 95, 95, 100	60.90 [60.26–64.10]
Static WH8, no replay	ClassIL	26, 26, 26, 26, 26	25.00 [25.00–25.00]
Static WH8 + replay	TaskIL	120, 121, 122, 121, 122	77.56 [76.92–78.21]
Static WH8 + replay	ClassIL	70, 64, 68, 67, 65	64.42 [61.54–67.31]
Legacy WH8, no replay	TaskIL	114, 119, 115, 117, 116	74.36 [73.08–76.28]
Legacy WH8, no replay	ClassIL	23, 39, 37, 34, 34	32.69 [22.12–37.50]
Legacy WH8 + replay	TaskIL	126, 124, 123, 129, 122	79.49 [78.21–82.69]
Legacy WH8 + replay	ClassIL	52, 47, 48, 49, 50	47.12 [45.19–50.00]
NN1, M2	TaskIL	138	88.46 [88.46–88.46]
NN1, M2	ClassIL	66	63.46 [63.46–63.46]

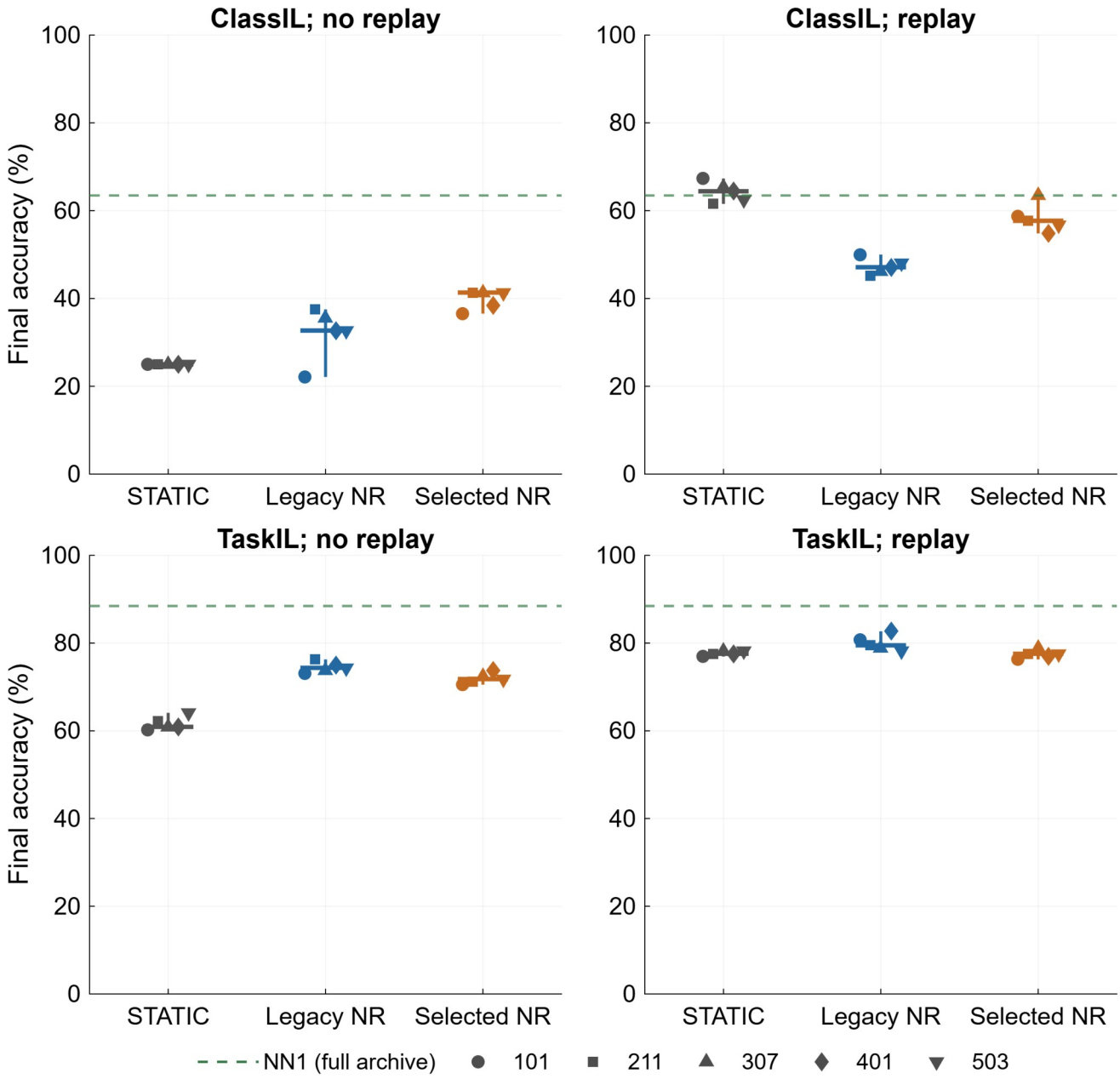


Figure S6. Final M2 accuracy for original static and legacy Walsh controls and the inner-selected procedure, with replay absent/present. All five numerical starts, median bars and minimum–maximum ranges are displayed. TaskIL is a separately trained transfer of the ClassIL-selected vector. The NN1 dashed line uses the full introduced TRAIN archive. Denominators are 104 ClassIL and 156 TaskIL views; history and tuning costs differ. Starts are not physical replications.

The selected no-replay ClassIL totals are 38/43/43/40/43, versus legacy 23/39/37/34/34. Under replay they are 61/60/66/57/59, versus legacy 52/47/48/49/50. The static replay counts 70/64/68/67/65 exceed the selected procedure in every paired seed, so selection is not described as improving on that control. The original nearest-record total is 66/104. Selected TaskIL totals are below both the legacy TaskIL median and the nearest-record total; these adverse transfer results are retained.

Table S9. Final old/new correct-count vectors for selected and legacy procedures. Five entries follow the same seed order as Table S8. ClassIL old/new denominators are 78/26; TaskIL old/new denominators are 104/52. Old and new totals add to each final total. Current acquisition, final retention and total accuracy should be interpreted together.

Procedure / replay	Stream	Earlier correct / N	Newest correct / N
Inner-selected WH8, no replay	ClassIL	34, 38, 40, 36, 36 / 78	4, 5, 3, 4, 7 / 26
Inner-selected WH8, no replay	TaskIL	79, 79, 79, 80, 78 / 104	31, 32, 34, 35, 34 / 52
Inner-selected WH8 + replay	ClassIL	38, 41, 45, 41, 37 / 78	23, 19, 21, 16, 22 / 26
Inner-selected WH8 + replay	TaskIL	80, 80, 83, 80, 81 / 104	39, 41, 40, 40, 40 / 52
Legacy WH8, no replay	ClassIL	11, 26, 27, 24, 26 / 78	12, 13, 10, 10, 8 / 26
Legacy WH8, no replay	TaskIL	79, 82, 79, 80, 78 / 104	35, 37, 36, 37, 38 / 52
Legacy WH8 + replay	ClassIL	45, 45, 46, 46, 45 / 78	7, 2, 2, 3, 5 / 26
Legacy WH8 + replay	TaskIL	89, 86, 86, 87, 85 / 104	37, 38, 37, 42, 37 / 52

Without replay, selected ClassIL new-class performance has median 4/26, below the legacy median 10/26, while earlier correct counts rise to median 36/78 from 26/78. With replay, the selected new-class median is 21/26 versus legacy 3/26, while the earlier median is 41/78 versus 45/78. These are observed acquisition–retention redistributions, not a single monotone retention benefit. Every stage's old/new, class-specific and TaskIL-pair counts is retained in the machine-readable companion tables.

Actual computation, reuse and technical history

Table S10. Actual accepted activity by phase. A trajectory is a logical three-stage procedure; exact shared stages and whole historical trajectories can be reused without new optimization. Actual updates equal 600 times newly fitted stages. Prediction rows are computational views, not specimens. Producer/owned times are sums of accepted batch measurements, not complete end-to-end elapsed time or per-method latency.

Phase / logical work	New fits / updates	New / reused predictions	Producer / owned seconds
coarse 440 trajectories; 1320 stage views	980 / 588,000	41,184 / 0	122.817 / 359.921
refine 16 trajectories; 48 stage views	40 / 24,000	1,476 / 0	5.330 / 21.234
confirm 300 trajectories; 900 stage views	810 / 486,000	18,720 / 0	82.631 / 234.297
outer_selected 100 trajectories; 300 stage views	130 / 78,000	3,150 / 2,310	105.723 / 185.735

The accepted execution comprises 62 batches and 856 logical trajectories: 440 coarse, 16 refinement, 300 confirmation and 100 final outer trajectories. These represent 2,568 logical stage instances, but exact stage reuse and historical imports reduce new work to 1,960 fits and 1,176,000 optimizer updates. Of 66,840 logical prediction rows, 64,530 are newly computed and 2,310 reuse historical decisions from 40 complete outer trajectories. Summed accepted producer time is 316.501 seconds and owned-batch wall time 801.187 seconds. Synthetic control work is excluded from these real-source totals. No Fisher computation or physical validation is added.

A technical stop after the first 16 accepted outer trajectories occurred while importing a verified empty new-cache artifact. The next attempted batch stopped before any trajectory or fit; its separate recorded duration was 9.219 seconds. No scientific coefficients or model code were altered. A metadata-only empty-cache adapter

repair preserved provenance, and a separately frozen 84-job continuation completed the exact remaining outer set. The composite acceptance retains both histories and all original accepted artifacts. The failed attempt is not included among the 62 accepted batches or charged as new optimization.

The initial controls, checkpoint/gradient/state inspections and independent saved-output audits check mathematical implementation, membership and count consistency. They do not rerun every optimizer step in an independent framework. Scalar array payloads, saved file bytes, process commit and measured wall time remain distinct; shared stages do not establish intrinsic algorithmic efficiency. Newly selected Walsh states are not automatically included in the older Simulink verification or public v30.0 package.

Complete development evidence

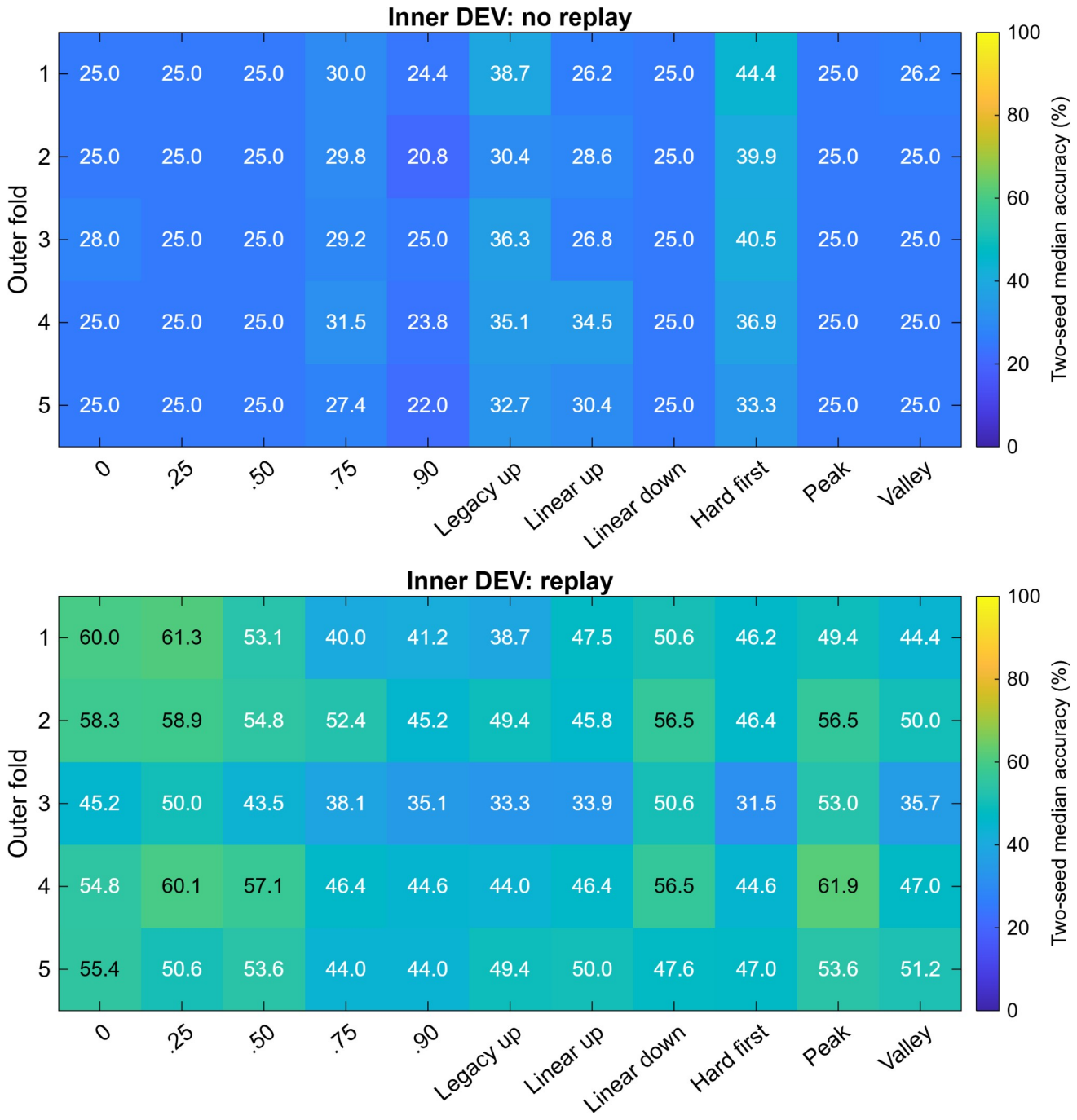


Figure S7. Complete coarse inner-development ClassIL results for the eleven declared vectors. Cells show the two-start median percentage after pooling each candidate's two inner whole-group folds; the same two starts are reused across overlapping development allocations. Rows identify outer folds, columns identify coarse candidate vectors and the two panels separate replay absent/present. No cell displays outer-test accuracy. N is 80 per start in outer fold 1 and 84 otherwise. The optional midpoint vectors and confirmation counts are reported separately in Tables S11–S12.

Table S11. Confirmation counts for every screened finalist. Each vector pools three inner whole-group folds. Five counts follow starts 101/211/307/401/503; N is 80 for outer fold 1 and 84 otherwise. The selected tag follows confirmation ranks alone. The two finalists in each context are compared with the frozen ranking rule; outer outcomes play no part.

Outer / replay	Candidate	DEV correct counts / N	Decision
1 / No	UP_NR_LEGACY	22, 35, 32, 33, 27 / 80	Selected
1 / No	HARD_FIRST_THEN_NR	27, 27, 24, 29, 32 / 80	Other finalist
1 / Yes	F00_STATIC	43, 43, 50, 42, 46 / 80	Selected
1 / Yes	F25	43, 40, 43, 43, 43 / 80	Other finalist
2 / No	HARD_FIRST_THEN_NR	39, 34, 36, 32, 38 / 84	Selected
2 / No	R2_MIX	32, 28, 39, 34, 29 / 84	Other finalist
2 / Yes	F00_STATIC	54, 53, 53, 53, 55 / 84	Selected
2 / Yes	F25	50, 51, 51, 52, 51 / 84	Other finalist
3 / No	HARD_FIRST_THEN_NR	33, 26, 30, 37, 35 / 84	Selected
3 / No	UP_NR_LEGACY	30, 35, 36, 28, 29 / 84	Other finalist
3 / Yes	PEAK	45, 52, 44, 48, 50 / 84	Selected
3 / Yes	DOWN_LINEAR	51, 43, 46, 45, 52 / 84	Other finalist
4 / No	HARD_FIRST_THEN_NR	39, 37, 37, 40, 36 / 84	Selected
4 / No	UP_NR_LEGACY	22, 28, 33, 30, 23 / 84	Other finalist
4 / Yes	F25	46, 46, 47, 48, 45 / 84	Selected
4 / Yes	PEAK	48, 45, 49, 45, 43 / 84	Other finalist
5 / No	HARD_FIRST_THEN_NR	33, 32, 36, 33, 40 / 84	Selected
5 / No	UP_NR_LEGACY	22, 30, 34, 29, 24 / 84	Other finalist
5 / Yes	F00_STATIC	44, 43, 44, 44, 42 / 84	Selected
5 / Yes	F50	38, 36, 39, 32, 36 / 84	Other finalist

Table S12. Complete coarse and refinement DEV count vectors. Counts follow screen starts 101/211 and pool the two whole-group inner folds. The same denominator applies to each seed. Every coarse candidate appears in every context; only qualified contexts add R1/R2. Complete all-stage job counts and exact frozen identities remain in the accompanying CSV/JSON records.

Outer / replay	Candidate, rank order	DEV correct, 101/211	N per seed
1 / No	HARD_FIRST_THEN_NR	36, 35	80
1 / No	UP_NR_LEGACY	29, 33	80
1 / No	R2_MIX	31, 25	80

Outer / replay	Candidate, rank order	DEV correct, 101/211	N per seed
1 / No	F75	19, 29	80
1 / No	R1_CONST	20, 23	80
1 / No	UP_LINEAR	20, 22	80
1 / No	VALLEY	20, 22	80
1 / No	F00_STATIC	20, 20	80
1 / No	F25	20, 20	80
1 / No	F50	20, 20	80
1 / No	PEAK	20, 20	80
1 / No	DOWN_LINEAR	20, 20	80
1 / No	F90	9, 30	80
1 / Yes	F25	51, 47	80
1 / Yes	F00_STATIC	47, 49	80
1 / Yes	F50	45, 40	80
1 / Yes	DOWN_LINEAR	43, 38	80
1 / Yes	PEAK	41, 38	80
1 / Yes	UP_LINEAR	37, 39	80
1 / Yes	HARD_FIRST_THEN_NR	38, 36	80
1 / Yes	VALLEY	37, 34	80
1 / Yes	F90	31, 35	80
1 / Yes	F75	32, 32	80
1 / Yes	UP_NR_LEGACY	29, 33	80
2 / No	HARD_FIRST_THEN_NR	36, 31	84
2 / No	R2_MIX	35, 27	84
2 / No	UP_NR_LEGACY	24, 27	84
2 / No	F75	21, 29	84
2 / No	UP_LINEAR	27, 21	84
2 / No	F00_STATIC	21, 21	84
2 / No	F25	21, 21	84
2 / No	F50	21, 21	84

Outer / replay	Candidate, rank order	DEV correct, 101/211	N per seed
2 / No	R1_CONST	21, 21	84
2 / No	PEAK	21, 21	84
2 / No	DOWN_LINEAR	21, 21	84
2 / No	VALLEY	21, 21	84
2 / No	F90	8, 27	84
2 / Yes	F25	53, 46	84
2 / Yes	F00_STATIC	49, 49	84
2 / Yes	DOWN_LINEAR	46, 49	84
2 / Yes	PEAK	50, 45	84
2 / Yes	F50	44, 48	84
2 / Yes	F75	42, 46	84
2 / Yes	VALLEY	42, 42	84
2 / Yes	UP_NR_LEGACY	41, 42	84
2 / Yes	HARD_FIRST_THEN_NR	40, 38	84
2 / Yes	UP_LINEAR	40, 37	84
2 / Yes	F90	37, 39	84
3 / No	HARD_FIRST_THEN_NR	26, 42	84
3 / No	UP_NR_LEGACY	25, 36	84
3 / No	F75	21, 28	84
3 / No	F00_STATIC	26, 21	84
3 / No	UP_LINEAR	21, 24	84
3 / No	F25	21, 21	84
3 / No	F50	21, 21	84
3 / No	PEAK	21, 21	84
3 / No	DOWN_LINEAR	21, 21	84
3 / No	VALLEY	21, 21	84
3 / No	F90	12, 30	84
3 / Yes	PEAK	44, 45	84
3 / Yes	DOWN_LINEAR	43, 42	84

Outer / replay	Candidate, rank order	DEV correct, 101/211	N per seed
3 / Yes	F25	42, 42	84
3 / Yes	F00_STATIC	38, 38	84
3 / Yes	F50	39, 34	84
3 / Yes	F75	33, 31	84
3 / Yes	VALLEY	29, 31	84
3 / Yes	F90	31, 28	84
3 / Yes	UP_LINEAR	31, 26	84
3 / Yes	UP_NR_LEGACY	30, 26	84
3 / Yes	HARD_FIRST_THEN_NR	27, 26	84
4 / No	HARD_FIRST_THEN_NR	28, 34	84
4 / No	UP_NR_LEGACY	28, 31	84
4 / No	UP_LINEAR	28, 30	84
4 / No	F75	20, 33	84
4 / No	F00_STATIC	21, 21	84
4 / No	F25	21, 21	84
4 / No	F50	21, 21	84
4 / No	PEAK	21, 21	84
4 / No	DOWN_LINEAR	21, 21	84
4 / No	VALLEY	21, 21	84
4 / No	F90	18, 22	84
4 / Yes	PEAK	52, 52	84
4 / Yes	F25	50, 51	84
4 / Yes	F50	48, 48	84
4 / Yes	DOWN_LINEAR	47, 48	84
4 / Yes	F00_STATIC	48, 44	84
4 / Yes	VALLEY	40, 39	84
4 / Yes	F75	39, 39	84
4 / Yes	UP_LINEAR	38, 40	84
4 / Yes	F90	37, 38	84

Outer / replay	Candidate, rank order	DEV correct, 101/211	N per seed
4 / Yes	HARD_FIRST_THEN_NR	37, 38	84
4 / Yes	UP_NR_LEGACY	38, 36	84
5 / No	HARD_FIRST_THEN_NR	31, 25	84
5 / No	UP_NR_LEGACY	23, 32	84
5 / No	UP_LINEAR	26, 25	84
5 / No	F75	17, 29	84
5 / No	F00_STATIC	21, 21	84
5 / No	F25	21, 21	84
5 / No	F50	21, 21	84
5 / No	PEAK	21, 21	84
5 / No	DOWN_LINEAR	21, 21	84
5 / No	VALLEY	21, 21	84
5 / No	F90	7, 30	84
5 / Yes	F00_STATIC	50, 43	84
5 / Yes	F50	45, 45	84
5 / Yes	PEAK	45, 45	84
5 / Yes	VALLEY	44, 42	84
5 / Yes	F25	41, 44	84
5 / Yes	UP_LINEAR	42, 42	84
5 / Yes	UP_NR_LEGACY	41, 42	84
5 / Yes	DOWN_LINEAR	39, 41	84
5 / Yes	HARD_FIRST_THEN_NR	43, 36	84
5 / Yes	F75	34, 40	84
5 / Yes	F90	34, 40	84

The retained report tables are NR_final_all_seed_metrics_v1.csv (62 rows), NR_final_median_min_max_v1.csv (14 rows), NR_selected_ten_vectors_v1.csv (10 choices), NR_stage_old_new_counts_v1.csv (186 stage rows), NR_class_counts_v1.csv and NR_TaskIL_pair_counts_v1.csv. The complete development files retain all-stage job counts, final-job counts and pooled candidate counts; none is an added physical observation. Their accepted handoff, selection receipts, input/code pins and independent final report audit bind the displayed tables to the frozen computational record.

Scope of the inference

The selected ClassIL improvement is relative to the legacy vector, not to the static replay control or every baseline. Confirmation may choose static targets. The family is finite, candidate generation is nonrecursive, and the other continual-learning families receive no matched tuning budget. Stage-wise alpha changes are not an adaptive learning-rate controller. The separate TaskIL transfer, all starts, unsuccessful candidates and reduced newest-class acquisition without replay remain visible. Nested group separation controls the stated computational access but does not erase prior inspection of this small archive or establish new-specimen performance.

S10 Separate endpoint-inclusive NR response

S10.1 Fixed family and common evaluation

This response analysis is separate from the inner-selected procedure in S9. The family contains six constant NR vectors—0, 0.25, 0.5, 0.75, 0.9 and 1 in every stage—and two stage schedules, increasing (0,0.5,1) and decreasing (1,0.5,0). NR is the target-mixing alpha, not the optimizer learning rate. Rates remain fixed within a learning stage. The eight vectors were declared before this response run; the report retains every setting rather than selecting an outer-query winner. The primary seven-profile taskwise comparison is identified in Section S10.4 and excludes only the additional constant NR = 0.9 control.

The original modulus/hardness inputs, five whole-group folds, stage-1 TRAIN-only normalization, Walsh eight-bit codebook and five starts (101, 211, 307, 401, 503) remain fixed. Each trajectory uses the same 52-parameter network and 600 Adam updates per stage, including carried parameters, moments and update count. TaskIL supplies its eligible pair and updates one network sequentially across the three tasks; ClassIL trains a separate trajectory with its expanding label set. The final TaskIL population is 156 decision views, including repeated PLA+ views; the ClassIL population is 104 records. Their differing arrival structures prevent attributing the whole stream contrast solely to task identity.

Without replay, no earlier exemplar is stored for rehearsal. With replay, the deterministic buffer retains two rows per introduced class, at most eight rows. Stage-3 eligible earlier rows are four in TaskIL and six in ClassIL after the declared current-anchor filtering. Every replay target remains a hard class code. The nearest-record reference instead retains all 80 or 84 available training records at the final stage. Rate-response comparisons therefore do not establish equal-memory superiority.

S10.2 Endpoint meaning and acquisition–retention interpretation

The stage target is $q = (1 - NR) \times \text{hard code} + NR \times \text{cached incoming prediction}$. The cached sigmoid prediction is detached and fixed for current TRAIN rows throughout the stage. At NR = 0 the target is the hard code. At NR = 1 the current hard-code contribution is absent, and the initial current-data BCE gradient is zero because the prediction equals its cached target. L2 weight regularization, carried Adam moments and hard replay targets can nevertheless move the model during the 600 updates. Later current-data gradients can also oppose departures from the cache. Stage-1 self-targets come from an untrained network. NR = 1 is consequently a self-target endpoint, not a frozen-parameter control.

With replay, constant NR = 0.5 gives the largest TaskIL median in this family (82.69%), whereas NR = 0 gives the largest ClassIL median (64.42%). Constant NR = 1 with replay gives ClassIL earlier-class and newest-class medians of 58.97% and 0%; static targets give 56.41% and 88.46%. Without replay, constant NR = 1 has newest-class correct-count vector [1, 9, 1, 0, 5]/26, so zero acquisition is not universal across endpoint contexts. The increasing ClassIL schedule has zero final PLA-CF recognition in all five starts with either memory setting. TaskIL's newest task includes both PLA+ and PLA-CF and must not be confused with new-class acquisition; for example, increasing NR with replay has a 50.00% newest-task median despite class-4 counts [0, 0, 0, 1, 0]/26.

Tables S13–S14 preserve all final seed counts and their old/new components. Panel (c) of Main Table 8 gives the seven-profile Walsh percentages; the additional 0.9 control remains here. Section S12 and panels (a)–(b) of Main Table 8 add the matched uncoded and ordinary-coded results. The starts describe numerical initialization variation, not independent specimens or confidence intervals. The response family provides no continuous optimum, significance test, unseen production test or equal search budget for EWC/LwF. The earlier S9 selection and its transfer results remain unchanged.

S10.3 Computational scope and retained evidence

The family contains 800 logical trajectories and 2,400 logical stages: eight vectors × two memory settings × two streams × five folds × five starts. Exact reuse avoids repeating accepted work: 110 complete trajectories

are copied (100 historical static controls and ten prior constant-0.25 trajectories), with 600 further stage aliases. The remaining 1,470 newly fitted stages use 882,000 Adam updates. The saved report contains 43,680 query decisions, comprising 37,695 newly evaluated and 5,985 copied decisions. These counts are distinct from the earlier 856-trajectory inner-selection analysis in S9; neither analysis creates another physical sample population.

The local computational extension v3 preserves the earlier A31 additions and the broader eight-profile Walsh protocol, code, states, decisions, all-stage/class summaries, figures and audit records. That broader family's original handoff used saved numerical checks without added Simulink verification. The matched seven-profile WH8 subset is subsequently included with NL4 and OH8 in the frozen-model inference and confidence-threshold count verification in Section S12.4. The additional NR = 0.9 history and the inner-selected sensitivity remain outside that matched scope. The retained response figures use saved report values without regenerating model predictions.

S10.4 Taskwise NR modes and primary seven-profile subset

NR is fixed during each task's 600 updates and may change only between tasks. A complete three-task sequence is one experiment. The three modes are constant, (c, c, c) for c = 0, 0.25, 0.5, 0.75 or 1; increasing, (0, 0.5, 1); and decreasing, (1, 0.5, 0). These seven profiles identify 700 logical trajectories already contained in the broader 800-trajectory response family. The subset is defined by the stated profiles, not by performance, and requires no refitting or additional forward predictions.

The subset contains 2,100 stages and 38,220 saved all-stage decisions; 18,200 decisions are final-stage views. Tables S13 (TaskIL) and S14 (ClassIL) retain every five-start total/earlier/current count vector. The primary comparison uses all their rows except the two constant NR = 0.9 rows. Figures S8–S9 and the broader family's original activity counts in Section S10.3 are retained. A five-task sequence such as (0, 0.25, 0.5, 0.75, 1) illustrates the increasing mode; the corresponding A31 sequence has three genuine tasks, with NR values (0, 0.5, 1).

Table S13. All five final TaskIL correct-count vectors for the fixed eight-vector family. Entries follow starts 101 / 211 / 307 / 401 / 503; denominators are stated in the headers. The newest task is the PLA+/PLA-CF pair, not only the newest class. One network learns the three tasks sequentially.

NR stages 1 / 2 / 3 **Replay**	**Total correct** **N = 156**	**Earlier tasks** **N = 104**	**Newest task** **N = 52**
0 / 0 / 0 No	[94, 97, 95, 95, 100]	[52, 54, 52, 52, 55]	[42, 43, 43, 43, 45]
0 / 0 / 0 Yes	[120, 121, 122, 121, 122]	[78, 79, 80, 80, 80]	[42, 42, 42, 41, 42]
0.25 / 0.25 / 0.25 No	[102, 102, 104, 99, 104]	[59, 58, 61, 55, 61]	[43, 44, 43, 44, 43]
0.25 / 0.25 / 0.25 Yes	[120, 127, 124, 122, 126]	[78, 83, 81, 79, 83]	[42, 44, 43, 43, 43]
0.5 / 0.5 / 0.5 No	[119, 109, 107, 114, 115]	[74, 69, 64, 71, 71]	[45, 40, 43, 43, 44]
0.5 / 0.5 / 0.5 Yes	[132, 126, 123, 129, 130]	[90, 84, 82, 86, 88]	[42, 42, 41, 43, 42]
0.75 / 0.75 / 0.75 No	[114, 122, 119, 121, 108]	[73, 80, 76, 80, 66]	[41, 42, 43, 41, 42]
0.75 / 0.75 / 0.75 Yes	[123, 129, 128, 126, 122]	[84, 90, 86, 86, 85]	[39, 39, 42, 40, 37]
0.9 / 0.9 / 0.9 No	[86, 122, 115, 119, 91]	[56, 81, 78, 80, 54]	[30, 41, 37, 39, 37]
0.9 / 0.9 / 0.9 Yes	[119, 127, 120, 129, 117]	[80, 87, 83, 86, 81]	[39, 40, 37, 43, 36]
1 / 1 / 1 No	[32, 130, 100, 114, 35]	[25, 89, 71, 77, 15]	[7, 41, 29, 37, 20]
1 / 1 / 1 Yes	[65, 118, 96, 123, 66]	[53, 81, 72, 83, 46]	[12, 37, 24, 40, 20]
0 / 0.5 / 1 No	[107, 109, 108, 107, 108]	[81, 83, 82, 81, 82]	[26, 26, 26, 26, 26]
0 / 0.5 / 1 Yes	[109, 108, 109, 115, 113]	[83, 82, 83, 88, 87]	[26, 26, 26, 27, 26]
1 / 0.5 / 0 No	[93, 96, 94, 94, 95]	[51, 53, 51, 51, 52]	[42, 43, 43, 43, 43]
1 / 0.5 / 0 Yes	[128, 121, 125, 119, 125]	[83, 77, 80, 76, 80]	[45, 44, 45, 43, 45]

Table S14. All five final ClassIL correct-count vectors for the fixed eight-vector family. Entries follow starts 101 / 211 / 307 / 401 / 503; denominators are stated in the headers. The newest class is PLA-CF; earlier classes are PLA+, ABS and PETG.

NR stages 1 / 2 / 3 **Replay**	**Total correct** **N = 104**	**Earlier classes** **N = 78**	**Newest class** **N = 26**
0 / 0 / 0 No	[26, 26, 26, 26, 26]	[0, 0, 0, 0, 0]	[26, 26, 26, 26, 26]
0 / 0 / 0 Yes	[70, 64, 68, 67, 65]	[46, 41, 45, 44, 42]	[24, 23, 23, 23, 23]
0.25 / 0.25 / 0.25 No	[26, 26, 26, 26, 26]	[0, 0, 0, 0, 0]	[26, 26, 26, 26, 26]
0.25 / 0.25 / 0.25 Yes	[62, 61, 66, 67, 63]	[40, 42, 45, 49, 41]	[22, 19, 21, 18, 22]
0.5 / 0.5 / 0.5 No	[26, 26, 26, 26, 26]	[0, 0, 0, 0, 0]	[26, 26, 26, 26, 26]
0.5 / 0.5 / 0.5 Yes	[61, 63, 57, 53, 56]	[47, 49, 44, 43, 41]	[14, 14, 13, 10, 15]
0.75 / 0.75 / 0.75 No	[26, 34, 28, 23, 25]	[0, 17, 3, 2, 1]	[26, 17, 25, 21, 24]
0.75 / 0.75 / 0.75 Yes	[51, 54, 50, 51, 49]	[47, 52, 43, 45, 47]	[4, 2, 7, 6, 2]
0.9 / 0.9 / 0.9 No	[11, 38, 27, 15, 30]	[8, 27, 24, 14, 14]	[3, 11, 3, 1, 16]
0.9 / 0.9 / 0.9 Yes	[48, 50, 46, 45, 47]	[48, 50, 45, 45, 47]	[0, 0, 1, 0, 0]
1 / 1 / 1 No	[10, 45, 35, 24, 12]	[9, 36, 34, 24, 7]	[1, 9, 1, 0, 5]
1 / 1 / 1 Yes	[45, 47, 45, 49, 46]	[45, 47, 45, 49, 46]	[0, 0, 0, 0, 0]
0 / 0.5 / 1 No	[43, 35, 49, 44, 38]	[43, 35, 49, 44, 38]	[0, 0, 0, 0, 0]
0 / 0.5 / 1 Yes	[49, 50, 49, 49, 48]	[49, 50, 49, 49, 48]	[0, 0, 0, 0, 0]
1 / 0.5 / 0 No	[26, 26, 26, 26, 26]	[0, 0, 0, 0, 0]	[26, 26, 26, 26, 26]
1 / 0.5 / 0 Yes	[59, 57, 58, 67, 55]	[36, 36, 34, 44, 31]	[23, 21, 24, 23, 24]

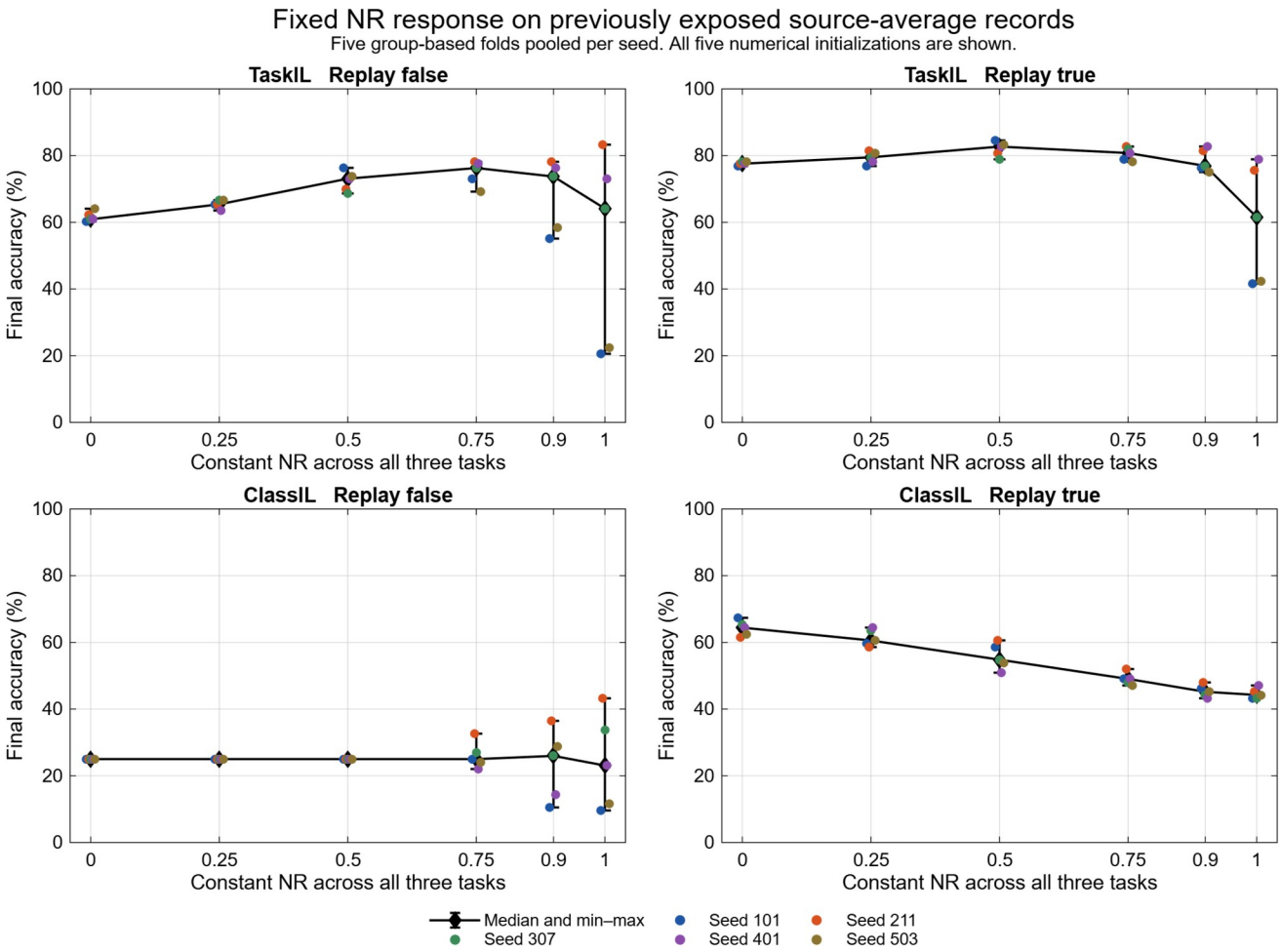


Figure S8. Final accuracy for the six prespecified constant target-mixing coefficients, with the same coefficient held across all three learning stages. Panels show TaskIL without/with replay and ClassIL without/with replay. Each coloured marker is one of the five numerical initializations after pooling the unchanged five held-out-group folds. Diamonds and vertical ranges show the median and minimum–maximum across those five starts. Small horizontal offsets separate overlapping seed markers and do not change the tested coefficient. Lines connect the tested settings as visual guides; they do not interpolate accuracy at untested rates. Final denominators are 156 TaskIL decision views and 104 ClassIL views from the same 104 previously exposed source averages. Starts are not independent physical samples.

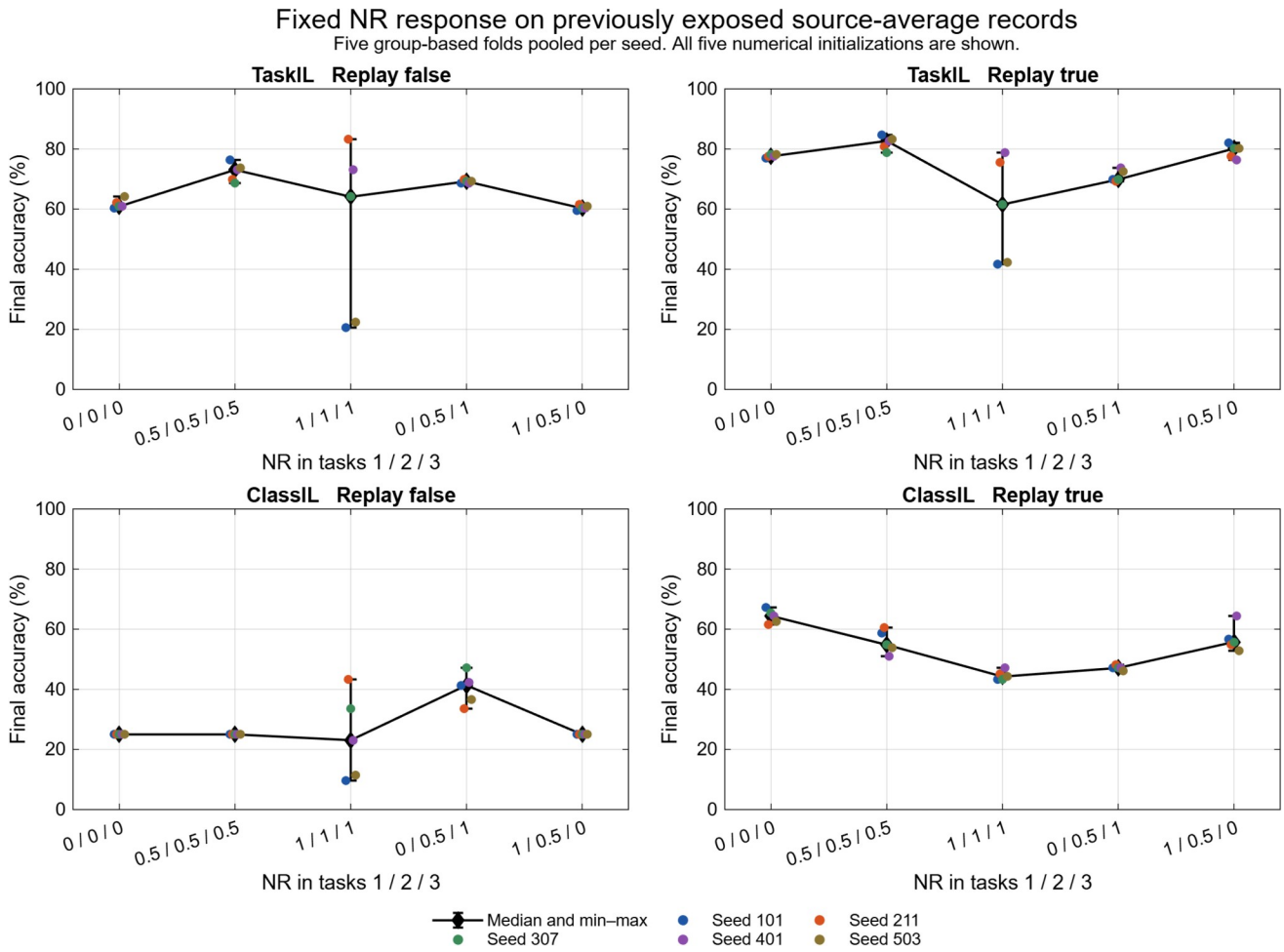


Figure S9. Final accuracy for three constant controls and the full rising and falling stage schedules: (0,0,0), (0.5,0.5,0.5), (1,1,1), (0,0.5,1) and (1,0.5,0). Coefficients change only between stages. One network is updated sequentially across all three stages of each TaskIL trajectory; ClassIL has its own separately trained trajectory. The four panels use the same stream/replay arrangement and five-start median/minimum–maximum display as the constant-coefficient figure. Marker offsets improve visibility, and connecting lines are categorical guides, not interpolation between untested rate schedules. TaskIL and ClassIL retain denominators of 156 and 104 decision views. The complete eight-vector family and every seed outcome are retained in the tables; no outer-result winner is selected.

S11 Reuse and scientific scope

Original MATLAB/helper code is distributed under MIT. Source-derived data and the authors' figures, documentation, computed results and model artifacts use CC BY 4.0 within the stated rights. Source data and the negotiated-method implementation retain their original notices and attribution. The file-specific scope and reproduction notes are included in the supplement.

The archived v30.0 materials remain at https://doi.org/10.5281/zenodo.23474663 and the matching GitHub release. This revision's additional figures and printing-input reference results, and its accepted finite Walsh sensitivity, are local A31 revision materials. They are not represented as already published in the v30.0 archive. The historical v27.0 material remains separate. The printing-input and inner-selected Walsh additions retain their saved-result checks; the matched three-family bank has the separate Simulink inference verification described in Section S12.4. These computational additions do not create a new blind dataset, identified specimens or physical validation.

The broader Walsh response in S10, its primary seven-profile subset, and the matched uncoded/ordinary-coded additions in S12 are supplied with the local A31 computational extension. The original eight-profile Walsh evidence remains intact; its seven-profile contribution is reused from saved results. The new uncoded and

ordinary-coded runs are accounted for separately in S12. These additions have not been deposited at the existing v30.0 DOI. Their numerical checks retain the distinction between saved consistency, archived Simulink inference and physical validation.

S12 Matched uncoded and coded taskwise negotiation

S12.1 Target definitions and shared information access

The matched response compares uncoded negotiated learning (NL4), ordinary-coded OH8 and retained Walsh-coded WH8. The uncoded model is the 32-parameter 2–4–4 tanh network used by the original cross-entropy references. Its current eligible-label probabilities use softmax at temperature 1. The task-entry teacher is detached: q = (1 − alpha) × one-hot label + alpha × incoming probability. Current cross-entropy is averaged over the current TRAIN rows. The teacher sees only those rows and the currently eligible labels; no DEV or outer query enters the target cache. This is an operational adaptation of the target-mixing idea to this source-record task; its fixed eligibility masks, temperature and task arrivals define the implementation being compared.

OH8 uses the 52-parameter 2–4–8 sigmoid network and codebook [I4, 0], with four learned zero-target bits. Its target is q = (1 − alpha) × hard code + alpha × incoming sigmoid prediction, with BCE averaged over current rows and all eight bits. WH8 retains the existing codebook and eight-bit implementation. The uncoded and coded procedures therefore differ in capacity, loss and decoder. OH8 and WH8 share architecture and loss but differ in bit density and distance geometry. This comparison does not attribute a difference solely to Walsh orthogonality.

A complete three-task sequence is one experiment. NR = alpha is constant for all 600 updates within a task and changes only at task boundaries. The seven profiles are F00 = (0,0,0), F25 = (0.25,0.25,0.25), F50 = (0.5,0.5,0.5), F75 = (0.75,0.75,0.75), F100 = (1,1,1), UP = (0,0.5,1) and DOWN = (1,0.5,0). These are five constant controls and two changing taskwise modes; the optimizer learning rate remains unchanged. The five whole-group folds, five starts (101, 211, 307, 401, 503), first-task TRAIN normalization, arrival rules and deterministic two-row-per-class replay buffer remain fixed. Each stream trains its own sequential trajectory. No outer-result selection or new inner tuning is performed.

Only current TRAIN targets are mixed. Replay uses hard labels or hard codes, with the original TaskIL admission mask where applicable; current-anchor filtering is unchanged. The incoming teacher and current target cache remain fixed through all 600 task updates. Parameters, first and second Adam moments and the update counter carry between tasks. At NR = 1 the current target contains no direct hard-label contribution. At task entry its data-loss gradient is zero up to numerical roundoff, while L2, replay and carried optimizer moments may still move the model. This is not a frozen-network control; first-task self-targets also do not represent previously learned classes.

S12.2 Complete final counts and acquisition–retention

Main Table 8 reports each family's median and minimum–maximum over five starts, after pooling the unchanged folds within each start. Tables S15–S18 retain every new uncoded/ordinary-coded final correct-count vector and its earlier/current components. Retained Walsh counts remain in Tables S13–S14; the seven-profile comparison uses all rows except the two NR = 0.9 rows. Full all-stage, material-specific and fold counts remain in the accompanying CSV files. TaskIL's newest unit is the PLA+/PLA-CF task, not only PLA-CF, whereas ClassIL's newest unit is the single PLA-CF class. The total TaskIL denominator of 156 repeats the PLA+ decision view across three pairs; it does not create 156 independent source records.

Table S15. NL4 final TaskIL correct-count vectors for all seven profiles and both memory settings. Entries follow starts 101 / 211 / 307 / 401 / 503 after pooling all five held-out-group folds. Profile definitions are given in Section S12.1; denominators are stated in the headers. The newest task contains PLA+ and PLA-CF, whereas the earlier two tasks each retain their declared pair.

Profile **Replay**	**Total correct** **N = 156**	**Earlier tasks** **N = 104**	**Newest task** **N = 52**
F00 No	[112, 117, 115, 114, 107]	[70, 74, 71, 71, 63]	[42, 43, 44, 43, 44]
F00 Yes	[112, 114, 117, 122, 124]	[69, 71, 72, 80, 76]	[43, 43, 45, 42, 48]
F25 No	[128, 122, 110, 121, 119]	[84, 77, 66, 78, 76]	[44, 45, 44, 43, 43]
F25 Yes	[106, 112, 99, 107, 115]	[62, 67, 58, 65, 71]	[44, 45, 41, 42, 44]
F50 No	[123, 120, 120, 115, 120]	[79, 77, 76, 73, 76]	[44, 43, 44, 42, 44]
F50 Yes	[99, 98, 109, 93, 109]	[61, 57, 66, 57, 65]	[38, 41, 43, 36, 44]
F75 No	[121, 109, 119, 120, 122]	[76, 69, 76, 75, 78]	[45, 40, 43, 45, 44]
F75 Yes	[86, 97, 81, 87, 88]	[60, 67, 55, 56, 60]	[26, 30, 26, 31, 28]
F100 No	[106, 114, 78, 47, 123]	[71, 84, 52, 19, 90]	[35, 30, 26, 28, 33]
F100 Yes	[77, 79, 78, 78, 81]	[52, 53, 52, 54, 53]	[25, 26, 26, 24, 28]
UP No	[118, 117, 113, 114, 114]	[83, 84, 78, 86, 76]	[35, 33, 35, 28, 38]
UP Yes	[114, 113, 114, 120, 125]	[77, 80, 78, 86, 86]	[37, 33, 36, 34, 39]
DOWN No	[112, 110, 116, 106, 113]	[69, 64, 72, 61, 68]	[43, 46, 44, 45, 45]
DOWN Yes	[121, 129, 130, 128, 132]	[78, 85, 86, 84, 88]	[43, 44, 44, 44, 44]

Table S16. NL4 final ClassIL correct-count vectors for all seven profiles and both memory settings. Entries follow starts 101 / 211 / 307 / 401 / 503 after pooling all five held-out-group folds. Profile definitions are given in Section S12.1; denominators are stated in the headers. The newest class is PLA-CF; earlier classes are PLA+, ABS and PETG.

Profile **Replay**	**Total correct** **N = 104**	**Earlier classes** **N = 78**	**Newest class** **N = 26**
F00 No	[26, 26, 26, 26, 28]	[0, 0, 0, 0, 2]	[26, 26, 26, 26, 26]
F00 Yes	[61, 62, 57, 60, 56]	[37, 41, 36, 37, 32]	[24, 21, 21, 23, 24]
F25 No	[26, 27, 26, 26, 25]	[0, 1, 0, 0, 0]	[26, 26, 26, 26, 25]
F25 Yes	[55, 62, 58, 56, 59]	[35, 44, 39, 37, 42]	[20, 18, 19, 19, 17]
F50 No	[26, 26, 26, 26, 26]	[0, 0, 0, 0, 0]	[26, 26, 26, 26, 26]
F50 Yes	[60, 59, 59, 60, 56]	[47, 47, 46, 45, 45]	[13, 12, 13, 15, 11]
F75 No	[26, 26, 26, 26, 26]	[0, 0, 0, 0, 0]	[26, 26, 26, 26, 26]
F75 Yes	[53, 55, 53, 53, 52]	[46, 50, 46, 49, 47]	[7, 5, 7, 4, 5]
F100 No	[44, 44, 15, 8, 45]	[44, 44, 13, 8, 43]	[0, 0, 2, 0, 2]
F100 Yes	[48, 45, 47, 45, 45]	[48, 45, 47, 45, 45]	[0, 0, 0, 0, 0]
UP No	[43, 46, 48, 44, 50]	[43, 46, 48, 44, 50]	[0, 0, 0, 0, 0]
UP Yes	[47, 43, 48, 48, 44]	[47, 43, 48, 48, 44]	[0, 0, 0, 0, 0]
DOWN No	[26, 26, 26, 26, 26]	[0, 0, 0, 0, 0]	[26, 26, 26, 26, 26]
DOWN Yes	[58, 59, 61, 54, 57]	[37, 36, 38, 34, 35]	[21, 23, 23, 20, 22]

Table S17. OH8 final TaskIL correct-count vectors for all seven profiles and both memory settings. Entries follow starts 101 / 211 / 307 / 401 / 503 after pooling all five held-out-group folds. Profile definitions are given in Section S12.1; denominators are stated in the headers. The newest task contains PLA+ and PLA-CF, whereas the earlier two tasks each retain their declared pair.

Profile **Replay**	**Total correct** **N = 156**	**Earlier tasks** **N = 104**	**Newest task** **N = 52**
F00 No	[95, 95, 94, 96, 101]	[52, 52, 52, 52, 57]	[43, 43, 42, 44, 44]
F00 Yes	[129, 124, 126, 125, 128]	[87, 80, 82, 81, 86]	[42, 44, 44, 44, 42]
F25 No	[108, 103, 104, 102, 107]	[62, 59, 61, 58, 62]	[46, 44, 43, 44, 45]
F25 Yes	[129, 122, 124, 125, 132]	[85, 80, 81, 81, 88]	[44, 42, 43, 44, 44]
F50 No	[110, 114, 112, 107, 110]	[66, 71, 70, 62, 66]	[44, 43, 42, 45, 44]
F50 Yes	[128, 129, 126, 123, 130]	[84, 87, 83, 80, 87]	[44, 42, 43, 43, 43]
F75 No	[115, 112, 110, 109, 117]	[75, 73, 70, 69, 75]	[40, 39, 40, 40, 42]
F75 Yes	[128, 132, 124, 124, 135]	[89, 91, 86, 85, 90]	[39, 41, 38, 39, 45]
F100 No	[126, 105, 75, 56, 122]	[91, 91, 57, 47, 88]	[35, 14, 18, 9, 34]
F100 Yes	[117, 97, 87, 80, 114]	[81, 82, 74, 66, 80]	[36, 15, 13, 14, 34]
UP No	[109, 110, 110, 109, 107]	[83, 84, 84, 83, 81]	[26, 26, 26, 26, 26]
UP Yes	[115, 115, 113, 111, 118]	[89, 87, 86, 84, 91]	[26, 28, 27, 27, 27]
DOWN No	[98, 95, 95, 93, 95]	[53, 53, 52, 50, 52]	[45, 42, 43, 43, 43]
DOWN Yes	[128, 123, 122, 125, 126]	[84, 79, 79, 81, 83]	[44, 44, 43, 44, 43]

Table S18. OH8 final ClassIL correct-count vectors for all seven profiles and both memory settings. Entries follow starts 101 / 211 / 307 / 401 / 503 after pooling all five held-out-group folds. Profile definitions are given in Section S12.1; denominators are stated in the headers. The newest class is PLA-CF; earlier classes are PLA+, ABS and PETG.

Profile **Replay**	**Total correct** **N = 104**	**Earlier classes** **N = 78**	**Newest class** **N = 26**
F00 No	[26, 26, 26, 26, 26]	[0, 0, 0, 0, 0]	[26, 26, 26, 26, 26]
F00 Yes	[63, 64, 66, 69, 62]	[42, 43, 44, 46, 40]	[21, 21, 22, 23, 22]
F25 No	[26, 26, 26, 26, 26]	[0, 0, 0, 0, 0]	[26, 26, 26, 26, 26]
F25 Yes	[58, 61, 60, 62, 68]	[38, 43, 40, 43, 46]	[20, 18, 20, 19, 22]
F50 No	[26, 26, 26, 26, 26]	[0, 0, 0, 0, 0]	[26, 26, 26, 26, 26]
F50 Yes	[61, 59, 57, 64, 58]	[49, 48, 46, 49, 46]	[12, 11, 11, 15, 12]
F75 No	[26, 34, 26, 27, 25]	[0, 21, 18, 19, 0]	[26, 13, 8, 8, 25]
F75 Yes	[50, 48, 52, 50, 54]	[45, 46, 49, 47, 48]	[5, 2, 3, 3, 6]
F100 No	[39, 53, 24, 29, 41]	[21, 52, 23, 28, 41]	[18, 1, 1, 1, 0]
F100 Yes	[48, 47, 47, 48, 46]	[45, 47, 47, 48, 46]	[3, 0, 0, 0, 0]
UP No	[30, 44, 45, 47, 48]	[30, 44, 45, 47, 48]	[0, 0, 0, 0, 0]
UP Yes	[50, 51, 47, 52, 53]	[50, 51, 47, 52, 53]	[0, 0, 0, 0, 0]
DOWN No	[26, 26, 26, 26, 26]	[0, 0, 0, 0, 0]	[26, 26, 26, 26, 26]
DOWN Yes	[56, 56, 56, 57, 52]	[32, 33, 34, 34, 30]	[24, 23, 22, 23, 22]

With replay and constant NR = 1, ClassIL earlier-class medians are 57.69% for NL4, 60.26% for OH8 and 58.97% for WH8; newest-class medians are 0% for all three. The static-control newest-class medians are 88.46%, 84.62% and 88.46%. These are context-specific results with replay, rather than a claim that every NR = 1 context has zero acquisition. All starts and both memory settings remain visible in the tables. No profile exceeds the 88.46% TaskIL nearest-record reference, which retains all available training rows; replay retains at most eight rows. The increasing ClassIL profile gives 0/26 correct final PLA-CF decisions in all five starts, for every family and both memory settings (30 contexts). Replay is not uniformly helpful: at constant NR = 0.5, uncoded TaskIL medians are 76.92% without replay and 63.46% with replay.

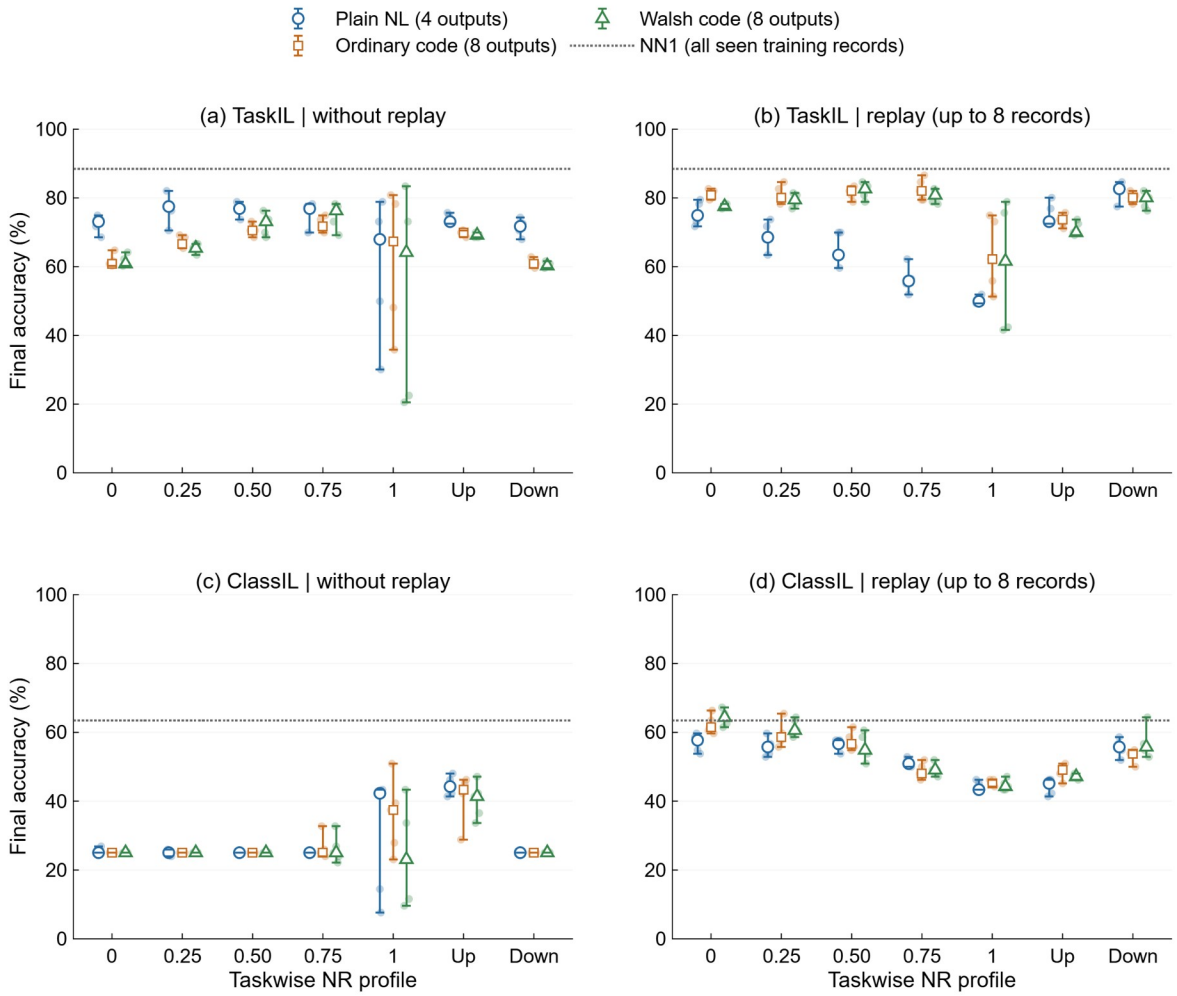


Figure S10. Final accuracy across seven taskwise negotiation-rate profiles. Symbols give the five-start median; whiskers show minimum to maximum and faint dots retain every start. The first five profiles keep NR at 0, 0.25, 0.5, 0.75 or 1 in all three tasks. Up uses (0, 0.5, 1); Down uses (1, 0.5, 0). NR never changes within a task. Each start pools five whole-group outer folds. NN1 retains all introduced training records, whereas replay retains at most eight. Final populations are 156 TaskIL views reusing 104 source averages and 104 ClassIL records. Numerical starts are not physical replications. NL4 and the coded models differ in width, capacity, loss and decoder; ordinary and Walsh codes also differ in bit density and geometry. All profiles are reported without outer-query selection. Previously accepted Walsh results are reused.

S12.3 Actual computation, exact reuse and numerical checks

The new NL4/OH8 family contains 1,400 logical trajectories and 4,200 logical stages. Two hundred complete static trajectories are copied from accepted historical results. A further 1,025 stage instances use exact same-context numerical cache aliases. The remaining 2,575 newly fitted stages use 1,545,000 Adam updates. The two new families contribute 76,440 all-stage decisions: 65,520 newly evaluated decisions and 10,920 copied historical decisions. New teacher evaluation covers 110,240 TRAIN rows. Logical requests and reused payloads are therefore distinct from actual new fits and forward evaluations.

The retained seven-profile WH8 contribution contains 700 logical trajectories, 2,100 stages and 38,220 saved all-stage decisions. It adds no new fitting, optimizer updates, teacher evaluation or query prediction in the present matched report. The complete three-family comparison consequently comprises 2,100 logical trajectories, 6,300 stages and 114,660 all-stage decision views. The corresponding 3,780,000 logical optimizer updates are not all newly executed updates. The broader eight-profile Walsh activity in Section S10.3 remains unchanged; its historical execution costs are not presented as fresh matched-analysis cost or intrinsic standalone efficiency.

Every source batch was accepted through an independent saved-output check of information access, target caches, optimizer checkpoints, model states, predictions and count tables. A separate raw-decision reaggregation checked the 114,660 saved decision rows and all 84 family/stream/memory/profile summaries. It made no new experimental predictions or fits. These checks establish numerical consistency of the recorded procedure. The separate Simulink audit below verifies frozen inference against the original saved scientific outputs, including the confidence-gap groups used in selective-coverage reporting.

S12.4 Frozen-inference verification and exact coverage counts

The native Simulink component review covers the matched seven-profile NL4, OH8 and WH8 bank: 2,100 logical trajectories and 6,300 stage instances across both streams, both memory settings, five folds and five starts. Its 114,660 logical query views refer to the same published-average population. The native model receives saved network parameters, initial-TRAIN normalizers, eligible labels and the appropriate output codebook. It reproduces frozen network inference and decoding; it does not perform training, teacher construction or optimizer updates.

The earlier compiled-bank review retained 21,580 logical views from 14,560 distinct archived cases. Its 93,080 remaining views comprised 82,160 distinct new native cases and 10,920 restored duplicates, with 3,850 distinct numerical states. Reuse required exact model/input/mask/codebook identity; duplicate views added no physical observations. The updated NL4 route below recomputes the union of archived and new queries.

Independent comparisons check normalized inputs, logits, applicable probabilities or code outputs, decoder values, eligible masks, predicted labels, tie/validity status and decision counts. Undefined masked outputs and non-applicable distances retain their missing-value meaning. The numerical tolerance is 2×10^{-11} absolute plus 2×10^{-11} times the absolute reference value. Across the independent audit's finite scalar comparisons, the largest observed absolute difference was 4.974e-14. Source-row identity, candidate memberships, forced-choice labels and counts, and tie/validity status agree exactly. Native SLX/MAT banks, deployment data, case-to-source mappings and independent audit records accompany the local computational extension.

The normal-mode NL4 execution matches the original saved scientific reference in exact confidence-gap block membership and order, together with every C/W/A/F/N cut sequence. Its 32,240 distinct queries restore 38,220 logical views across 2,100 populations. Combined with the accepted compiled OH8/WH8 evidence, the audit covers all 6,300 populations and 114,660 views. The earlier compiled NL4 route's two near-tie partition differences are retained in the historical audit.

The NL4 model uses a normal-mode desktop MATLAB-engine call to the unchanged original decoder. Running it requires the accompanying external MATLAB helper and compatible MATLAB/Simulink. This route verifies frozen desktop simulation with an explicit MATLAB dependency; standalone generated-code evidence remains confined to the separately stated compiled scope. The verification covers the matched seven-profile bank. Earlier NR = 0.9 runs, inner-selected trajectories and printing-input references retain their separate evidence. The archived original M2 verification also remains separately identified. Physical specimen and production-workflow validation remain outside this computational verification.

S12.5 Interpretation and file scope

The matched comparison keeps the source rows, group folds and visible information fixed while testing a finite set of complete rate sequences. It does not establish a continuous optimum, equal model capacity, equal target entropy, a tuned comparison against EWC/LwF or equal memory with the full-history nearest-record rule. The same previously exposed 104 published averages remain the source population; starts are numerical initialization variation, not physical replication. The original material-confusion and printing-input interpretations remain in the main text and preceding sections.

The accompanying local computational extension supplies the matched protocol and implementation, fixed job membership, saved states and caches, decision CSVs, derived tables, plot sources and audit records. These local additions are separate from the established v30.0 GitHub/Zenodo archive. They retain MIT code terms, CC BY 4.0 scientific-material scope within the stated rights, and source/method notices. They do not assign a new public DOI or imply that the v30.0 archive already contains the new matched runs. Table S19 supplies the paired TaskIL confusion counts for the printing-input comparison.

S12.6 TaskIL confusion counts for the printing-input comparison

Table S19 reports the final TaskIL decisions for each of the three candidate material pairs. The two counts in each true-class column are listed in the order of the two labels in the task heading. Each pair contributes 52 decision views. PLA+ recurs across tasks; the 156 TaskIL views do not represent 156 distinct source averages.

Table S19. TaskIL confusion counts by task pair, deterministic classifier and input set. For the first and second true classes, the two counts give predictions in the ordered pair shown in the task label; diagonal counts are correct. M2 uses tensile modulus and Shore D hardness. M2P2 adds layer height and infill density.

Task and ordered class pair	Classifier	Input set	True first class → predicted [first, second]	True second class → predicted [first, second]	Correct / 52
Task 1: PLA+ / ABS	Nearest-record (NN1)	M2	23, 3	1, 25	48/52
Task 1: PLA+ / ABS	Nearest-record (NN1)	M2P2	26, 0	2, 24	50/52
Task 1: PLA+ / ABS	Nearest-centroid (PROTO)	M2	21, 5	3, 23	44/52
Task 1: PLA+ / ABS	Nearest-centroid (PROTO)	M2P2	21, 5	3, 23	44/52
Task 2: PLA+ / PETG	Nearest-record (NN1)	M2	22, 4	3, 23	45/52
Task 2: PLA+ / PETG	Nearest-record (NN1)	M2P2	25, 1	0, 26	51/52
Task 2: PLA+ / PETG	Nearest-centroid (PROTO)	M2	23, 3	4, 22	45/52
Task 2: PLA+ / PETG	Nearest-centroid (PROTO)	M2P2	23, 3	4, 22	45/52
Task 3: PLA+ / PLA-CF	Nearest-record (NN1)	M2	23, 3	4, 22	45/52
Task 3: PLA+ / PLA-CF	Nearest-record (NN1)	M2P2	25, 1	3, 23	48/52
Task 3: PLA+ / PLA-CF	Nearest-centroid (PROTO)	M2	23, 3	10, 16	39/52
Task 3: PLA+ / PLA-CF	Nearest-centroid (PROTO)	M2P2	23, 3	10, 16	39/52